Chapter 13

Boundary Value Problems via Fourier Series

13.1 Equilibria of Fisher with Neumann boundary conditions

Consider the one-dimensional parabolic equation

$$\begin{aligned} u_t &= u_{yy} + \lambda u(1-u), \qquad \lambda \in \mathbb{R} \\ u &= u(t,y), \quad y \in [0,\pi], \quad \frac{\partial u}{\partial y}(t,0) = \frac{\partial u}{\partial y}(t,\pi) = 0. \end{aligned} \tag{13.1}$$

Equation (13.1) is often called *Fisher's equation*.

Solutions of (13.1) can be expressed via the Fourier expansion

$$u(t,y) = \sum_{k=-\infty}^{\infty} a_k(t) e^{iky} = a_0(t) + 2\sum_{k=1}^{\infty} a_k(t)\cos(ky), \tag{13.2}$$

where $a_k \in \mathbb{R}$ and $a_{-k} = a_k$. Plugging (13.2) in (13.1) results in the infinite set of ODEs given by

$$\dot{a}_k = f_k(a) \stackrel{\text{def}}{=} \mu_k a_k - \lambda \sum_{\substack{k_1+k_2=k \\ k_1,k_2 \in \mathbb{Z}}} a_{k_1} a_{k_2}, \tag{13.3}$$

where

$$\mu_k \stackrel{\text{def}}{=} \lambda - k^2 \tag{13.4}$$

is the eigenvalue of the linear part of (13.1). Since $a_{-k} = a_k$, then $f_{-k} = f_k$. This implies that one can consider only the variables a_k for $k \geq 0$ and the functions f_k for $k \geq 0$. This is the reason why in this case we are going to use the one-sided sequences ℓ^1_ν. Note that looking for equilibria of the Fisher PDE (13.1) is equivalent to compute solutions of $f(a) = 0$ in ℓ^1_ν, for some $\nu > 1$ small enough.

13.2 Equilibria of the Chafee-Infante equation

Consider the parabolic partial differential equation (PDE)

$$u_t = u_{xx} - \lambda u(u^2 - 1) \tag{13.5}$$

defined on the domain $x \in [-1, 1]$. This equation is referred to by several different names depending on the community and the boundary conditions; with Dirichlet boundary conditions $u(t, -1) = u(t, 1) = 0$ it is often called the Chafee-Infante equation [7], while with Neuman boundary conditions $u_x(t, -1) = u_x(t, 1) = 0$ is typically called the Allen-Cahn equation [1]. As in the case of ordinary differential equations the first step in understanding the dynamics is to identify the equilibria. In the case of the Dirichlet boundary conditions this is equivalent to solving the system of ODEs

$$\begin{aligned} u_x &= v \\ v_x &= \lambda u(u^2 - 1) \end{aligned} \tag{13.6}$$

with the conditions that $u(-1) = 0$ and $u(1) = 0$. Since the additional conditions are associated with the boundary conditions of the PDE, it makes sense to refer to this as a boundary value problem (BVP). The phase portrait for this system is shown in Figure 13.1 and makes it clear that it is possible for the desired solution to exist. In fact, it is possible to have multiple solutions. As is demonstrated in [10, Section 3.6.1] for any fixed $\lambda > 0$ there exists a finite number of solutions however the number of solutions becomes unbounded as $\lambda \to \infty$. It should be noted that the technique of proof relies heavily on the fact that (13.6) is a two-dimensional Hamiltonian system and hence solutions to the BVP must be defined in terms of periodic solutions. If we consider higher order equations, then the argument described in [10] no longer applies. In particular, there is no reason a priori to believe that the solution to the BVP should lie on a periodic solution.

13.3 Equilibria of Swift-Hohenberg with Neumann boundary conditions

Let us consider the Swift-Hohenberg PDE with even periodic boundary conditions

$$\begin{aligned} &u_t = (\lambda - 1)u - 2u_{yy} - u_{yyyy} - u^3 , \quad \text{in } \Omega = [0, \frac{2\pi}{L}] \\ &u(t, y) = u(t, y + 2\pi/L) \ , \ u(t, y) = u(t, -y) \ , \quad \text{on } \partial\Omega. \end{aligned} \tag{13.7}$$

This model was originally introduced to describe the onset of Rayleigh-Bénard heat convection [37], where L is a fundamental wave number for the system size $\frac{2\pi}{L}$. The parameter λ corresponds to the Rayleigh number and its increase is associated with the appearance of multiple solutions that exhibit complicated patterns. For the computations presented

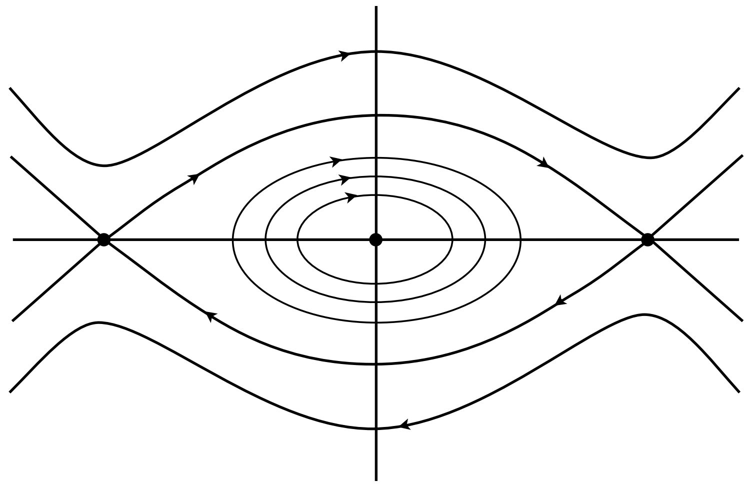

Figure 13.1: Phase portrait for (13.6).

here we fixed $L = 0.65$. At $\lambda = \left(1 - 4L^2\right)^2$, there is a pitchfork bifurcation from $u \equiv 0$. The bifurcating solution corresponds to the solution $\cos(2Ly)$.

Solutions of (13.7) can be expressed via the Fourier expansion

$$u(t,y) = \sum_{k=-\infty}^{\infty} a_k(t) e^{ikLy} = a_0 + 2\sum_{k=1}^{\infty} a_k(t)\cos(kLy), \tag{13.8}$$

where $a_k \in \mathbb{R}$ and $a_{-k} = a_k$. Plugging (13.8) in (13.7) results in the infinite set of ODEs given by

$$\dot{a}_k = F_k(a) \stackrel{\text{def}}{=} \mu_k a_k - \sum_{k_1+k_2+k_3=k} a_{k_1} a_{k_2} a_{k_3}, \tag{13.9}$$

where

$$\mu_k \stackrel{\text{def}}{=} \lambda - \left(1 - k^2L^2\right)^2 \tag{13.10}$$

is the eigenvalue of the linear part of (13.7). Since $a_{-k} = a_k$, then $F_{-k} = F_k$. This implies that one can consider only the variables a_k for $k \geq 0$ and the functions F_k for $k \geq 0$. This is the reason why in this case we are going to use the one-sided sequences ℓ^1_ν. Note that looking for equilibria of the Swift-Hohenberg PDE (13.7) is equivalent to compute solutions of $F(a) = 0$ in ℓ^1_ν, for some $\nu > 1$ small enough.

We fix $L = 0.65$ and leave λ as a continuation parameter. Equilibria $u = u(y)$ of (13.7) correspond to solutions of $F(a) = 0$, where $a = (a_k)_{k\geq 0}$ is the infinite sequence of Fourier coefficients and $F = (F_k)_{k\geq 0}$ is given component-wise by (13.9). In this case, the expansion (13.8) reads as

$$u(y) = \sum_{k\in\mathbb{Z}} a_k \cos(kLy) = a_0 + 2\sum_{k\geq 1} a_k \cos(kLy). \tag{13.11}$$

Since there is a natural symmetry $a_{-k} = a_k$ in the Fourier coefficients of u, we slightly adjust the definition of the space as follows: given an infinite sequence $a = (a_k)_{k\geq 0}$, define

$$\|a\|_\nu \stackrel{\text{def}}{=} \sum_{k\geq 0} |a_k|\nu^k$$

and the function space consisting of one-sided sequences

$$\ell^1_\nu = \{a = (a_k)_{k\geq 0} \;:\; \|a\|_\nu < \infty\}.$$

Endow the one-sided ℓ^1_ν with the following *extended* discrete convolution product: given $a = (a_k)_{k\geq 0}, b = (b_k)_{k\geq 0} \in \ell^1_\nu$, extend them with the symmetry $a_{-k} = a_k$ and $b_{-k} = b_k$, and define $a * b$ component-wise by

$$(a*b)_k \stackrel{\text{def}}{=} \sum_{\substack{k_1+k_2=k \\ k_1,k_2\in\mathbb{Z}}} a_{k_1} b_{k_2}, \quad k \geq 0.$$

Then

$$\|a*b\|_\nu \leq 4\|a\|_\nu \|b\|_\nu, \tag{13.12}$$

as can be checked by direct computation.

Given an infinite dimensional vector $v = (v_k)_{k\geq 0}$, we use the notation $v_F = (v_0, v_1, \ldots, v_{m-1}) \in \mathbb{R}^m$ to denote its finite dimensional projection. A Galerkin projection $F^{(m)} : \mathbb{R}^m \to \mathbb{R}^m$ is defined by $F^{(m)}(a_F) = (F(a_F, 0))_F$. Assume that using Newton's method, we computed a solution $\bar{a} = (\bar{a}_0, \bar{a}_1, \ldots, \bar{a}_{m-1})$ such that $F^{(m)}(\bar{a}) \approx 0$. Consider A_m computed so that $A_m \approx \left(DF^{(m)}(\bar{a})\right)^{-1}$ and assume that A_m is invertible. Recalling (13.10), set

$$A \stackrel{\text{def}}{=} \begin{bmatrix} A_m & & 0 & \\ & \mu_m^{-1} & & \\ 0 & & \mu_{m+1}^{-1} & \\ & & & \ddots \end{bmatrix}.$$

Take the Galerkin projection dimension m large enough so that $|\mu_k| \geq |\mu_m|$ for all $k \geq m$. Recalling (7.9), set

$$K \stackrel{\text{def}}{=} \max_{0\leq n\leq m-1} \frac{1}{\nu^n} \sum_{\ell=0}^{m-1} |(A_m)_{\ell,n}|\nu^\ell,$$

and define

$$\alpha_\nu = \max(K, \frac{1}{\mu_m}). \tag{13.13}$$

Then we have that

$$\|A\|_{\ell^1_\nu} \leq \alpha_\nu. \tag{13.14}$$

Proposition 13.3.1. *Define the Newton-like operator by*

$$T(a) = a - AF(a). \tag{13.15}$$

Then $T : \ell^1_\nu \to \ell^1_\nu$.

The goal is to demonstrate that nearby the approximate solution $\bar{a}$, there exists an exact solution of $F(a) = 0$, with F given component-wise by (13.9). In this case, F is defined on the function space ℓ^1_ν, hence there will be only one radii polynomial.

13.3.1 Computation of the radii polynomial

Let us now compute the bounds Y and Z in the context of the equilibria of the Swift-Hohenberg PDE (13.7). Recall that by considering a Galerkin projection $F^{(m)} : \mathbb{R}^m \to \mathbb{R}^m$, one computed a solution $\bar{a} = (\bar{a}_0, \bar{a}_1, \ldots, \bar{a}_{m-1})$ such that $F^{(m)}(\bar{a}) \approx 0$. Hence,

$$\left|[T(\bar{a}) - \bar{a}]_k\right| = \left|[-AF(\bar{a})]_k\right| \leq Y_k. \tag{13.16}$$

Compute $Y_F = (Y_0, Y_1, \ldots, Y_{m-1})^T$ with the formula

$$Y_F = |A_m(F(\bar{a}))_F|. \tag{13.17}$$

Since $\bar{a}_k = 0$ for all $k \geq m$, then $F_k(\bar{a}) = \mu_k \bar{a}_k - (\bar{a} * \bar{a} * \bar{a})_k = 0$ for all $k \geq 3m-2$. For $k = m, \ldots, 3m-3$, set

$$Y_k = \left| -\frac{1}{\mu_k} F_k(\bar{a}) \right| = \frac{1}{|\mu_k|} \left| (\bar{a} * \bar{a} * \bar{a})_k \right|. \tag{13.18}$$

Using (13.17) and (13.18), one may obtain

$$\|Y\|_\nu = \sum_{k=0}^{m-1} \left|[A_m(F(\bar{a}))_F]_k\right| \nu^k + \sum_{k=m}^{3m-3} \frac{1}{|\mu_k|} \left|(\bar{a} * \bar{a} * \bar{a})_k\right| \nu^k. \tag{13.19}$$

To simplify the computation of the bounds $Z_k(r)$ define the operator

$$A^\dagger \stackrel{\text{def}}{=} \begin{bmatrix} DF^{(m)}(\bar{a}) & & 0 & \\ & \mu_m & & \\ 0 & & \mu_{m+1} & \\ & & & \ddots \end{bmatrix}.$$

Considering $b, c \in B(r)$ and recalling the definition of the Newton-like operator (13.15), notice that

$$DT(\bar{a}+b)c = [I - ADF(\bar{a}+b)]c = [I - AA^\dagger]c - A[DF(\bar{a}+b)c - A^\dagger c]. \tag{13.20}$$

We now bound the $\nu-$norm of each of the terms in the right hand side of (13.20). Consider $u, v \in B(1)$ such that $b = ur$ and $c = vr$. Let

$$Z^{(0)} \stackrel{\text{def}}{=} \max_{0 \le n \le m-1} \frac{1}{\nu^n} \sum_{\ell=0}^{m-1} \left| \left(I_m - A_m Df^{(m)}(\bar{x}) \right)_{\ell,n} \right| \nu^\ell. \tag{13.21}$$

By definition of the diagonal tails of A and $A^\dagger$, the diagonal tail of $I - AA^\dagger$ is zero. Hence,

$$\|[I - AA^\dagger]c\|_\nu \le Z^{(0)} r.$$

The next step is to bound the term $\| - A[DF(\bar{x} + b)c - A^\dagger c]\|_\nu$. For this, notice that for $k = 0, \dots, m-1$,

$$\begin{aligned}
[DF(\bar{a} + b)c - A^\dagger c]_k &= \mu_k c_k - 3[(\bar{a} + b)^2 c]_k - [DF^{(m)}(\bar{a}) c_F]_k \\
&= \mu_k c_k - 3[(\bar{a} + b)^2 c]_k - (\mu_k c_k - 3[\bar{a}^2 c_F]_k) \\
&= -3[\bar{a}^2 c_I]_k - 6[\bar{a} b c]_k - 3[b^2 c]_k \\
&= \left(-3[\bar{a}^2 v_I]_k\right) r + \left(-6[\bar{a} u v]_k\right) r^2 + \left(-3[u^2 v]_k\right) r^3,
\end{aligned}$$

where

$$\begin{aligned}
v_I &\stackrel{\text{def}}{=} (0, 0, \dots, 0, v_m, v_{m+1}, \dots) \\
[\bar{a}^2 v_I]_k &= \sum_{\substack{k_1+k_2+k_3=k \\ |k_3| \ge m}} \bar{a}_{k_1} \bar{a}_{k_2} v_{k_3} \\
[\bar{a} u v]_k &= \sum_{k_1+k_2+k_3=k} \bar{a}_{k_1} u_{k_2} v_{k_3} \\
[u^2 v]_k &= \sum_{k_1+k_2+k_3=k} u_{k_1} u_{k_2} v_{k_3}.
\end{aligned}$$

Similarly, for $k \ge m$,

$$\begin{aligned}
[DF(\bar{a} + b)c - A^\dagger c]_k &= \mu_k c_k - 3[(\bar{a} + b)^2 c]_k - \mu_k c_k \\
&= \left(-3[\bar{a}^2 v]_k\right) r + \left(-6[\bar{a} u v]_k\right) r^2 + \left(-3[u^2 v]_k\right) r^3.
\end{aligned}$$

Using (13.12) and (13.14),

$$\begin{aligned}
\|A[DF(\bar a+b)c-A^\dagger c]\|_\nu &= \sum_{k=0}^{m-1}\left|\left(A_m[\left(-3[\bar a^2 v_I]_F\right)r+\left(-6[\bar a uv]_F\right)r^2+\left(-3[u^2v]_F\right)r^3]\right)_k\right|\nu^k \\
&\quad+\sum_{k\ge m}\left|\frac{1}{\mu_k}\left(-3[\bar a^2 v]_k r-6[\bar a uv]_k r^2-3[u^2v]_k r^3\right)\right|\nu^k \\
&\le \left(3\sum_{k=0}^{m-1}\left|\left(|A_m|[|\bar a|^2|v_I|]_F\right)_k\right|\nu^k+\frac{3}{|\mu_m|}\sum_{k\ge m}\left|[\bar a^2 v]_k\right|\nu^k\right)r \\
&\quad+6\|A\|_{\ell^1_\nu}\|\bar a*u*v\|_\nu r^2+3\|A\|_{\ell^1_\nu}\|u^2*v\|_\nu r^3 \\
&\le Z^{(1)}r+6\cdot 16\alpha_\nu\|\bar a\|_\nu r^2+3\cdot 16\alpha_\nu r^3,
\end{aligned}$$

where the bound $Z^{(1)}$ can be obtained using Lemma 7.3.11. However, such bounds can be computationally expensive. One way to circumvent this issue is to use the following coarser (yet faster to compute!) bound.

$$\begin{aligned}
&\sup_{\|v\|_\nu\le 1}\left(3\sum_{k=0}^{m-1}\left|\left(|A_m|[|\bar a|^2|v_I|]_F\right)_k\right|\nu^k+\frac{3}{|\mu_m|}\sum_{k\ge m}\left|[\bar a^2 v]_k\right|\nu^k\right) \\
&\qquad\le 3\sum_{k=0}^{m-1}\left|\left(|A_m|[|\bar a|^2\tilde\omega]_F\right)_k\right|\nu^k+\frac{48\|\bar a\|_\nu^2}{|\mu_m|},
\end{aligned}$$

where

$$\tilde\omega \stackrel{\rm def}{=} (0,0,\dots,0,\nu^{-m},\nu^{-(m+1)},\dots,\nu^{-(3m-3)}).$$

The above bound is the worst case scenario, in the sense that each component $k\ge m$ of v_I is replaced by $\frac{1}{\nu^k}$. But this bound is much faster to compute as it requires the evaluation of only one convolution term.

Recall the definition of α_ν in (13.13) and set

$$\begin{aligned}
Z^{(1)} &\stackrel{\rm def}{=} 3\sum_{k=0}^{m-1}\left|\left(|A_m|[|\bar a|^2\tilde\omega]_F\right)_k\right|\nu^k+\frac{48\|\bar a\|_\nu^2}{|\mu_m|}, && (13.22)\\
Z^{(2)} &\stackrel{\rm def}{=} 96\alpha_\nu\|\bar a\|_\nu, && (13.23)\\
Z^{(3)} &\stackrel{\rm def}{=} 48\alpha_\nu. && (13.24)
\end{aligned}$$

Combining (13.21), (13.22), (13.23) and (13.24), we set

$$\|Z(r)\|_\nu \stackrel{\rm def}{=} Z^{(3)}r^3+Z^{(2)}r^2+(Z^{(1)}+Z^{(0)})r. \qquad (13.25)$$

Finally combining (13.19) and (13.25), one can define the radii polynomial by

$$\begin{aligned} p(r,\nu) &= \|Z(r)\|_\nu - r + \|Y\|_\nu \\ &= Z^{(3)}r^3 + Z^{(2)}r^2 + \left(Z^{(1)} + Z^{(0)} - 1\right) r \\ &\quad + \sum_{k=0}^{m-1} |[A_m f_F(\bar a)]_k|\, \nu^k + \sum_{k=m}^{3m-3} \frac{1}{|\mu_k|}\, |(\bar a * \bar a * \bar a)_k|\, \nu^k. \end{aligned} \tag{13.26}$$

Next, we show some results about rigorous computations of equilibria of (13.7) using the radii polynomial approach.

13.3.2 Validated numerics for equilibria of Swift-Hohenberg: existence, isolation, and domain of analyticity

Recalling the Swift-Hohenberg PDE (13.7), we fix the fundamental wave number $L = 0.65$ for the system size $\frac{2\pi}{L}$. As mentioned in the introduction there is a pitchfork bifurcation from $u \equiv 0$ at $\lambda = \left(1 - 4L^2\right)^2$ which corresponds to the solution $\cos(2Ly)$. Using a numerical continuation method based on a predictor corrector algorithm we computed numerical approximations for a long branch of equilibria and single out the parameter values of $\lambda = 1$, $\lambda = 10$ and $\lambda = 3.5 \times 10^8$ for rigorous validation. Then, we used a computer program in MATLAB to compute with interval arithmetics (again using INTLAB) the coefficients $p(r,1)$ as defined in (13.26). Define

$$\mathcal{I} = \mathcal{I}(\nu) \stackrel{\text{def}}{=} \{r > 0 \mid p(r,\nu) < 0\}. \tag{13.27}$$

We constructed $\mathcal{I}(1) \neq \emptyset$ as defined in (13.27). Then, we used a bisection algorithm to find the maximal $\nu = \nu_{\max}$ for which $\mathcal{I}(\nu_{\max}) \neq \emptyset$. We could therefore maximize the lower bound on the domain of analyticity of the spatially periodic solutions. At $\lambda = 1$, we obtained $\nu_{\max} = 2.249$ so that the function is analytic on a strip of width at least 1.2469. At $\lambda = 10$, we obtained $\nu_{\max} = 1.584$ and the width of the strip is at least 0.70762. At $\lambda = 3.5 \times 10^8$, we obtained $\nu_{\max} = 1.003$ and the width of the strip is at least 0.0046085.

λ	ν	m	$\|Y\|_\nu$	$Z^{(1)}$	$Z^{(2)}$	$\mathcal{I}(\nu)$
1	1	18	2.9936×10^{-13}	1.222×10^{-3}	250.8012	$[2.9972 \times 10^{-13} \;\; 0.0039689]$
1	2.249	18	1.7541×10^{-4}	0.014095	1368.2697	$[0.00032037 \;\; 0.00040009]$
10	1	31	5.4406×10^{-12}	3.4333×10^{-3}	151.3427	$[5.4594 \times 10^{-12} \;\; 6.5776 \times 10^{-3}]$
10	1.584	31	3.1532×10^{-4}	0.013617	742.2918	$[0.00053541 \;\; 0.00079332]$
3.5×10^8	1	2103	2.1516×10^{-4}	0.18899	0.73393	$[0.00026536 \;\; 1.1047]$
3.5×10^8	1.003	2103	1.6521×10^{-3}	0.13209	1.2908	$[0.0019089 \;\; 0.67048]$

Figure 13.2: Different data for the proofs.

Using a numerical continuation method based on a predictor corrector algorithm we continued to a solution at $\lambda = 3.5 \times 10^8$, and proved that near the numerical approximation there exists an exact solution. The proof used a Fourier approximation to $m = 2103$ modes. The ℓ^1 error between the approximate solution and the exact solution is smaller than $r = 2.6536 \times 10^{-4}$.

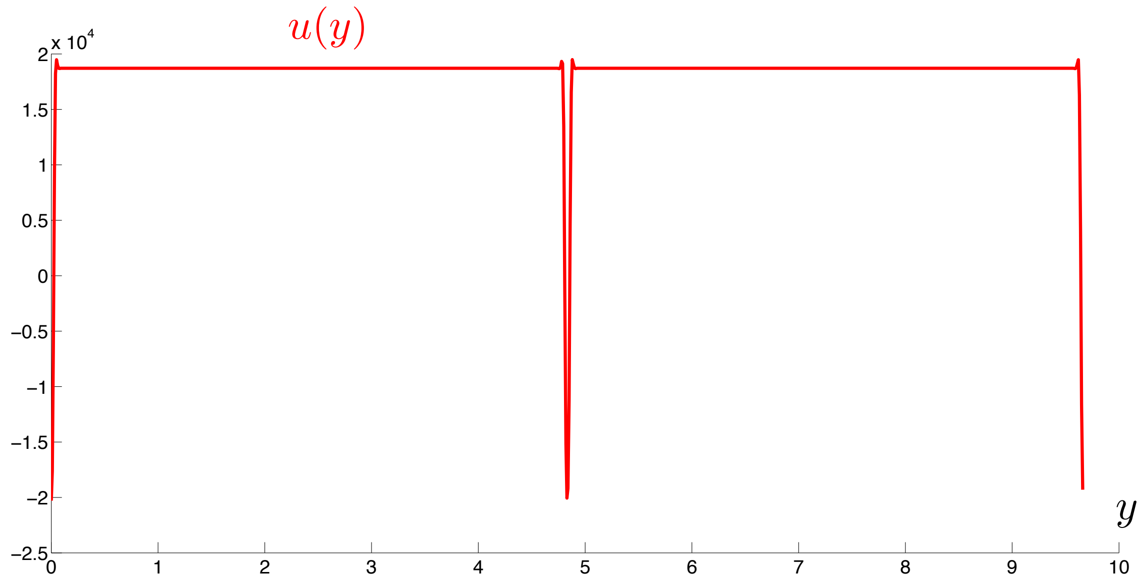

Figure 13.3: Equilibrium solution of (13.7) at $\lambda = 3.5 \times 10^8$. The computation is rigorously validated. The resulting error is smaller than the width of the curve. So we can say for example that the true solution exhibits the small spiking behaviour just before and just after the large spike, as shown in the figure. This phenomena is in this case not numerical error associated with the "Gibbs effect".

13.4 Equilibria of the Diblock-Copolymer equation

The diblock copolymer equation defined on the one-dimensional domain $\Omega = (0, \pi)$ is given by

$$u_t = F(\lambda, u) \stackrel{\text{def}}{=} -\left(u_{xx} + \lambda f(u)\right)_{xx} - \lambda\sigma(u - \mu) \ , \tag{13.28}$$

$$\text{subject to the mass constraint} \quad \mu = \frac{1}{\pi}\int_0^\pi u(t,x)dx \ ,$$

$$\text{as well as} \quad u_x(t,x) = u_{xxx}(t,x) = 0 \quad \text{for} \quad x = 0, \pi \quad \text{and for} \quad t > 0,$$

where in the following we use the nonlinearity $f(u) = u - u^3$. This partial differential equation is a model for microphase separation in diblock copolymers as described in [27]. In its original form it was proposed by Ohta and Kawasaki [28] and Bahiana and Oono [4], for more information regarding its derivation and the involved parameters we refer the reader to Choksi and Ren [12, 13].

Consider the steady states diblock copolymer equation (13.28) with $f(u) = u - u^3$ and $\mu = 0$, that is

$$F(\lambda, u) = -\left(u_{xx} + \lambda(u - u^3)\right)_{xx} - \lambda\sigma u = 0.$$

Plugging the cosine Fourier expansion

$$u(x) = \sum_{k\in\mathbb{Z}} a_k e^{ikx} = 2\sum_{k\geq 1} a_k \cos(kx), \quad (a_k \in \mathbb{R}, \quad a_{-k} = a_k, \quad a_0 = 0) \tag{13.29}$$

in $F(\lambda, u) = 0$ leads to solve $\sum_{k\in\mathbb{Z}} f_k e^{ikx} = 0$, with

$$f_k(a) \stackrel{\text{def}}{=} \mu_k(\lambda, \sigma) a_k - \lambda k^2 (a^3)_k, \tag{13.30}$$

where

$$\mu_k = \mu_k(\lambda, \sigma) \stackrel{\text{def}}{=} -k^4 + \lambda k^2 - \lambda\sigma, \tag{13.31}$$

and

$$(a^3)_k \stackrel{\text{def}}{=} \sum_{\substack{k_1+k_2+k_3=k \\ k_i\in\mathbb{Z}\setminus\{0\}}} a_{|k_1|} a_{|k_2|} a_{|k_3|}.$$

The relations $a_k \in \mathbb{R}$, $a_0 = 0$ and $a_{-k} = a_k$ imply the new relations $f_k \in \mathbb{R}$, $f_{-k} = f_k$ and $f_0 = 0$. Hence, we only need to solve $f_k = 0$ for $k \geq 1$. Set $f \stackrel{\text{def}}{=} (f_k)_{k\geq 1}$.

13.5 Equilibria of Gray-Scott

Chapter 14

Initial Value Problems via Chebyshev Series

The goal of this chapter is to show that the techniques developed in Chapter 7 and Chapter 10 can be applied to solve BVPs and in addition to characterize solutions to IVPs. On a mathematical level the only significant difference between the approach of this chapter and that of Chapter 10 is in the method of approximation. As is mentioned above solutions to general boundary value problems and clearly general solutions to initial value problems need not be periodic. Therefore, it is impossible to obtain point-wise convergence, as in Proposition 10.5.1, using Fourier series. With this in mind we turn to another set of orthogonal basis functions.

14.1 Chebyshev Polynomials

By Theorem **??**, Fourier modes form an orthogonal basis for $L^2([0,p])$ using the standard inner product. Changing the inner product will result in a different orthogonal basis. Using the inner product

$$(f,g) \stackrel{\text{def}}{=} \int_{-1}^{1} \frac{f(t)g(t)}{\sqrt{1-t^2}}\,dt$$

on the interval $[-1,1]$ leads to the following collection of functions.

Definition 14.1.1. The *Chebyshev polynomials* $T_k\colon [-1,1] \to \mathbb{R}$, $k = 0,1,2,\ldots$, are defined by $T_0(t) = 1$ and $T_1(t) = t$ and the recurrence relation

$$T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t), \quad k \geq 1.$$

Graphs of the Chebyshev polynomials $T_n(t)$, for $n = 0,\ldots,5$, are presented in Figure 14.1. As the following proposition shows Chebyshev polynomials are trigonometric functions *in disguise*.

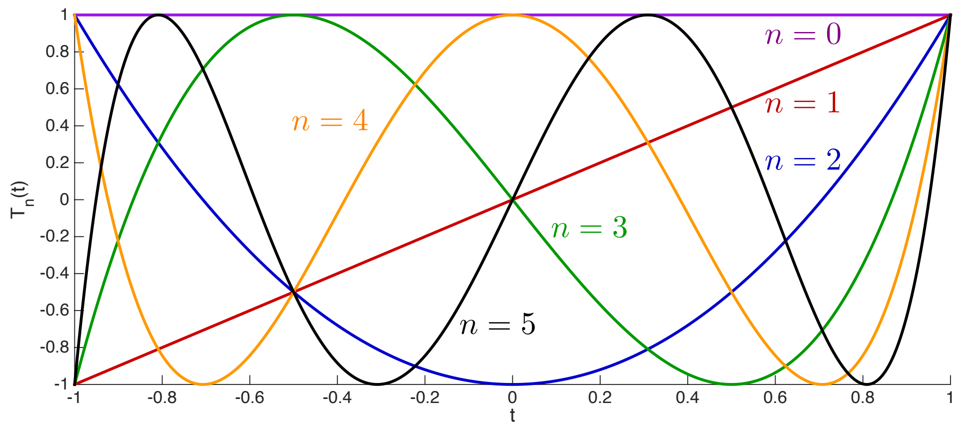

Figure 14.1: Chebyshev polynomials $T_n(t)$, for $n = 0, \dots, 5$.

Proposition 14.1.2. *For all $k \geq 0$, the Chebyshev polynomial T_k satisfies*

$$T_k(\cos\theta) = \cos(k\theta), \quad \forall \quad \theta \in [0, \pi]. \tag{14.1}$$

Proof. For $t \in [-1, 1]$, uniquely define $\theta = \cos^{-1}(t) \in [0, \pi]$ so that $t = \cos\theta$. Since $T_0(t) = 1$ and $T_1(t) = t$, then (14.1) is clearly true for $k = 0, 1$. Suppose (14.1) is true for all $k \leq n$. Then

$$\begin{aligned}
T_{n+1}(t) &= 2tT_n(t) - T_{n-1}(t) = 2\cos\theta\cos(n\theta) - \cos((n-1)\theta) \\
&= 2\cos\theta\cos(n\theta) - [\cos(n\theta)\cos(-\theta) - \sin(n\theta)\sin(-\theta)] \\
&= \cos\theta\cos(n\theta) - \sin(n\theta)\sin(\theta) \\
&= \cos((n+1)\theta). \quad \square
\end{aligned}$$

A direct calculation (using $t = \cos\theta$) shows that they are orthogonal, i.e.

$$\int_{-1}^{1} \frac{T_k(t)T_j(t)}{\sqrt{1-t^2}}\,dt = \int_0^{\pi} \cos(k\theta)\cos(j\theta)d\theta = \begin{cases} 0 & \text{if } j \neq k, \\ \pi & \text{if } j = k = 0, \\ \frac{\pi}{2} & \text{if } j = k > 0. \end{cases}$$

Furthermore, for every $k \geq 0$,

$$T_k(-1) = (-1)^k \quad \text{and} \quad T_k(1) = 1, \tag{14.2}$$

and

$$\begin{aligned}
&\int T_0(s)ds = s = T_1(s), \qquad \int T_1(s)ds = \frac{s^2}{2} = \frac{1}{4}(T_2(s) + T_0(s)), \\
&\int T_k(s)ds = \frac{1}{2}\left(\frac{T_{k+1}(s)}{k+1} - \frac{T_{k-1}(s)}{k-1}\right) + \text{constant}, \qquad k \geq 2.
\end{aligned} \tag{14.3}$$

Just as in the case of Fourier series, the Chebyshev polynomials form a basis for $L^2([-1,1])$. Thus given $u \in L^2([-1,1])$,

$$\lim_{K\to\infty}\left\|u(\cdot)-\sum_{k=0}^{K}\hat{u}(k)T_k(\cdot)\right\|_2=\lim_{K\to\infty}\left(\int_{-1}^{1}\frac{\left|u(t)-\sum_{k=0}^{K}\hat{u}(k)T_k(t)\right|^2}{\sqrt{1-t^2}}dt\right)^{\frac{1}{2}}=0$$

where

$$\hat{u}(k)\stackrel{\text{def}}{=}\frac{\alpha}{\pi}\int_{-1}^{1}\frac{u(t)T_k(t)}{\sqrt{1-t^2}}\,dt,\quad \alpha=\begin{cases}1 & \text{if } k=0\\ 2 & \text{if } k\geq 1,\end{cases} \tag{14.4}$$

are the *Chebyshev coefficients* of u. Because we are interested in solutions to ODEs our functions are differentiable and thus, as in the case of Fourier series, we can make use of stronger convergence results, as we see in Theorem 14.1.3. To prove the convergence, we recast the Chebyshev expansions in terms of Fourier series.

Theorem 14.1.3. *If $u\colon[-1,1]\to\mathbb{R}$ is real analytic function, then the Chebyshev series*

$$u(t)=\sum_{k\in\mathbb{N}}\hat{u}(k)T_k(t)=\sum_{k\in\mathbb{Z}}a_k\cos(k\theta)=\sum_{k\in\mathbb{Z}}a_ke^{ik\theta}$$

converges uniformly on $[-1,1]$. Moreover, there exist $C>0$ and $\nu>1$ such that

$$|\hat{u}(k)|\leq\frac{C}{\nu^k},\qquad \textit{for all } k\in\mathbb{N}.$$

Proof. Define $v\colon[0,2\pi]\to\mathbb{R}$ by

$$v(\theta)\stackrel{\text{def}}{=}u(\cos\theta). \tag{14.5}$$

This is a 2π-periodic real analytic function since u is real analytic. By the Paley-Wiener Theorem (Theorem 10.5.4), the Fourier expansion

$$v(\theta)=\sum_{k\in\mathbb{Z}}a_ke^{ik\theta}=\sum_{k\in\mathbb{Z}}a_k\cos(k\theta)$$

converges uniformly and $a\stackrel{\text{def}}{=}\{a_k\}_{k\in\mathbb{Z}}\in\ell^1_\nu$, for some $\nu>1$.

Using (14.1), we get for $k\geq 1$ that

$$\begin{aligned}\hat{u}(k)&=\frac{2}{\pi}\int_{-1}^{1}\frac{u(t)T_k(t)}{\sqrt{1-t^2}}\,dt\\&=\frac{2}{\pi}\int_{\pi}^{0}\frac{u(\cos\theta)T_k(\cos\theta)}{\sin\theta}\,(-\sin\theta)d\theta\\&=2\left(\frac{1}{2\pi}\int_0^{2\pi}v(\theta)\cos(k\theta)d\theta\right)\\&=2a_k.\end{aligned}$$

Similarly, we get that $\hat{u}(0)=a_0$, which proves the desired result. $\square$

The following result is the equivalent of Corollary **??** for non periodic problems.

Corollary 14.1.4. *Consider* $u = (u_1, u_2, \dots, u_n)^T$ *a function defined on the interval* $[-1, 1]$ *which is a solution of the analytic vector field*

$$\dot{u} = f(u). \tag{14.6}$$

Then each component $u_j : [-1, 1] \to \mathbb{R}$ *is analytic and therefore has a uniformly convergent series of the form* $u_j(t) = \sum_{k=0}^{\infty} \hat{u}_j(k) T_k(t)$. *For each* $j \in \{1, \dots, n\}$, *the sequence of Chebyshev coefficients* $\{\hat{u}_j(k)\}_{k \geq 0}$ *of* u_j *decreases to zero with a geometric decay. Moreover, there exist constants* $C < \infty$ *and* $\nu > 1$ *such that for all*

$$|\hat{u}_j(k)| \leq \frac{C}{\nu^k}, \qquad \textit{for all } k \geq 0 \textit{ and for all } j \in \{1, \dots, n\}.$$

Proof. By Proposition 19.6.7, u is a real analytic function. Hence, each component $u_j : \mathbb{R} \to \mathbb{R}$ is a real analytic function. By Theorem 14.1.3, there exists $\nu_j > 1$ and $C_j < \infty$ such that

$$|\hat{u}_j(k)| \leq \frac{C_j}{\nu_j^k}, \qquad \text{for all } k \in \mathbb{N}.$$

Letting

$$\nu \stackrel{\text{def}}{=} \min(\nu_1, \dots, \nu_1) > 1 \quad \text{and} \quad C \stackrel{\text{def}}{=} \max(C_1, \dots, C_n) < \infty,$$

the conclusion follows. $\square$

Consider two analytic functions $u_1, u_2 : [-1, 1] \to \mathbb{R}$. By Theorem 14.1.3, the function $u_j(t)$ $(j = 1, 2)$ can be expanded as a Chebyshev series

$$u_j(t) = \sum_{k \in \mathbb{N}} \hat{u}_j(k) T_k(t) = \sum_{k \in \mathbb{Z}} (a_j)_k \cos(k\theta) = \sum_{k \in \mathbb{Z}} (a_j)_k e^{ik\theta}$$

with $(a_j)_0 = \hat{u}_j(0)$, $(a_j)_k = \hat{u}_2(j)/2$ and $(a_j)_{-k} = (a_j)_k$.

Therefore, as in the case of Fourier series,

$$\begin{aligned}
u_1(t)u_2(t) &= \left(\sum_{k_1 \geq 0} \hat{u}_1(k_1) T_{k_1}(t) \right) \left(\sum_{k_2 \geq 0} \hat{u}_2(k_2) T_{k_2}(t) \right) \\
&= \left(\sum_{k_1 \in \mathbb{Z}} (a_1)_{k_1} e^{ik_1\theta} \right) \left(\sum_{k_2 \in \mathbb{Z}} (a_2)_{k_2} e^{ik_2\theta} \right) \\
&= \sum_{k \in \mathbb{Z}} (a_1 * a_2)_k e^{ik\theta} \\
&= (a_1 * a_2)_0 T_0(t) + 2 \sum_{k \geq 1} (a_1 * a_2)_k T_k(t)
\end{aligned}$$

where

$$(a_1 * a_2)_k = \sum_{k_1+k_2=k} (a_1)_{k_1}(a_2)_{k_2}.$$

More generally, for $u_j(t) = \sum_{k\in\mathbb{Z}}(a_j)_k T_k(t)$, $j = 1,\ldots,n$, one gets that

$$u_1(t)u_2(t)\cdots u_n(t) = \sum_{k\in\mathbb{Z}}(a_1 * a_2 * \cdots * a_n)_k T_k(t),$$

where

$$(a_1 * a_2 * \cdots * a_n)_k = \sum_{k_1+k_2+\cdots+k_n=k} (a_1)_{k_1}(a_2)_{k_2}\cdots(a_n)_{k_n}.$$

Similarly as in Chapter 7 and Chapter 10, the philosophy of the computational method is to rigorously compute solutions u of an (IVP) or a (BVP) associated to (14.6) by recasting them as solutions of an operator equation of the form

$$F(x) = 0 \tag{14.7}$$

where x is in a Banach space X consisting of product of the space ℓ^1_ν.

14.2 Initial Value Problem

The first class of problems we address in this chapter are initial value problems associated to the general analytic vector field (14.6). As in Lemma 6.1.1, a standard trick is to integrate (14.6) and to realize that finding a solution u to the initial value problem (IVP) $\dot{u} = f(u)$, $u(-1) = p \in \mathbb{R}^n$ is equivalent to finding a solution u of the integral equation

$$\mathcal{F}(u)(t) \stackrel{\text{def}}{=} p + \int_{-1}^{t} f(u(s))ds - u(t) = 0, \tag{14.8}$$

where we consider $t \in [-1, 1]$. Note that this is not a restriction since in the autonomous vector field (14.6), a re-scaling of time could be considered. The goal is to develop a rigorous computational method based on Chebyshev series to compute solutions of (14.8).

Consider a component-wise Chebyshev expansion of the j-th component of u and $f(u)$ given respectively by

$$u_j(t) = (a_j)_0 + 2\sum_{k\geq 1}(a_j)_k T_k(t), \tag{14.9}$$

and

$$f_j(u(t)) = (c_j)_0 + 2\sum_{k\geq 1}(c_j)_k T_k(t), \tag{14.10}$$

where $c_j = \{(c_j)_k\}_{k\in\mathbb{N}}$. Substituting (14.9) and (14.10) into (14.8), and using properties (14.2) and (14.3), one obtains

$$\begin{aligned}
\mathcal{F}_j(u)(t) &= p_j + \int_{-1}^{t} \left((c_j)_0 + 2\sum_{k\geq 1} (c_j)_k T_k(s) \right) ds - \left((a_j)_0 + 2\sum_{k\geq 1} (a_j)_k T_k(t) \right) \\
&= p_j + (c_j)_0(t+1) + 2(c_j)_1 \left(\frac{1}{4}(T_2(t) + T_0(t)) - \frac{1}{2} \right) - \left((a_j)_0 + 2\sum_{k\geq 1} (a_j)_k T_k(t) \right) \\
&\quad + 2\sum_{k\geq 2} (c_j)_k \left[\frac{1}{2} \left(\frac{T_{k+1}(t)}{k+1} - \frac{T_{k-1}(t)}{k-1} \right) - \frac{(-1)^k}{k^2-1} \right] \\
&= p_j + (c_j)_0(T_1(t)+1) + 2(c_j)_1 \left(\frac{T_2(t)}{4} - \frac{1}{4} \right) - \left((a_j)_0 + 2\sum_{k\geq 1} (a_j)_k T_k(t) \right) \\
&\quad + 2\sum_{k\geq 2} (c_j)_k \left[\frac{1}{2} \left(\frac{T_{k+1}(t)}{k+1} - \frac{T_{k-1}(t)}{k-1} \right) - \frac{(-1)^k}{k^2-1} \right] \\
&= (\tilde{F}_j)_0 + 2\sum_{k\geq 1} (\tilde{F}_j)_k T_k(t),
\end{aligned}$$

where

$$\begin{aligned}
(\tilde{F}_j(a))_0 &\stackrel{\text{def}}{=} p_j - (a_j)_0 + (c_j)_0 - \frac{(c_j)_1}{2} - 2\sum_{k\geq 2} \frac{(-1)^k}{k^2-1}(c_j)_k, \\
(\tilde{F}_j(a))_k &\stackrel{\text{def}}{=} \frac{(c_j)_{k-1} - (c_j)_{k+1}}{2k} - (a_j)_k, \ \ k \geq 1.
\end{aligned}$$

Denote $a = (a_1, a_2, \ldots, a_n)$ and define $F(a) = (F_1(a), \ldots, F_n(a))$ component-wise by

$$(F_j(a))_k \stackrel{\text{def}}{=} \begin{cases} p_j - (a_j)_0 - 2\displaystyle\sum_{n=1}^{\infty} (-1)^n (a_j)_n, & k = 0, \\ 2k(a_j)_k + (c_j)_{k+1} - (c_j)_{k-1}, & k \geq 1. \end{cases} \tag{14.11}$$

$F(a)$ given component-wise by (14.11) is called the *IVP-operator* and finding a solution u of (14.8) is equivalent to finding a zero of the IVP-operator. To see this, note that $(F_j(a))_k = -2k(\tilde{F}_j(a))_k$ for $k \geq 1$ and that if $2k(a_j)_k = (c_j)_{k-1} - (c_j)_{k+1}$ for all $k \geq 1$ then

$$p_j - (a_j)_0 + (c_j)_0 - \frac{(c_j)_1}{2} - 2\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2-1}(c_j)_n = p_j - (a_j)_0 - 2\sum_{n=1}^{\infty} (-1)^n (a_j)_n. \tag{14.12}$$

Hence, finding a solution u of the initial value problem (14.8) is equivalent to finding a zero of the IVP-operator in the Banach space

$$X = \left(\ell^1_\nu\right)^n.$$

Example 14.2.1. Consider the initial value problem

$$\dot{u} = u(u-1), \quad u(-1) = p \in (-1,1)$$

Hence, $f(u(t)) = u(u-1) = u^2 - u$. Considering a Chebyshev expansion of $u(t) = a_0 + 2\sum_{k\geq 1} a_k T_k(t)$, we get

$$f(u(t)) = \left(a_0 + 2\sum_{k\geq 1} a_k T_k(t)\right)^2 - \left(a_0 + 2\sum_{k\geq 1} a_k T_k(t)\right) = c_0 + 2\sum_{k\geq 1} c_k T_k(t),$$

where

$$c_k = (a*a)_k - a_k = \sum_{k_1+k_2=k} a_{k_1} a_{k_2} - a_k,$$

and so that the IVP-operator to solve is $F(a) = 0$ given component-wise by

$$F_k(a) \stackrel{\text{def}}{=} \begin{cases} p - a_0 - 2\sum_{n=1}^{\infty} (-1)^n a_n, & k = 0, \\ 2k a_k + (a*a)_{k+1} - a_{k+1} - (a*a)_{k-1} + a_{k-1}, & k \geq 1. \end{cases}$$

14.3 Exercise

1.

2. One dimensional Ginzburg-Landau model of superconductivity [16].

$$\begin{cases} \ddot{\phi} = k^2 \phi(\phi^2 + a^2 - 1) \\ \ddot{a} = \phi^2 a \\ \dot{\phi}\,(\pm d) = 0, \qquad \dot{a}\,(\pm d) = h_e \end{cases} \tag{14.13}$$

   Write an integral operator $\mathcal{F} : \mathbb{R}^2 \times (C^\infty[-1,1])^4 \to \mathbb{R}^2 \times (C^\infty[-1,1])^4$ such that solutions of $\mathcal{F}(\theta, u) = 0$ correspond to solutions of the BVP (14.13).

Chapter 15

Connecting Orbits

15.1 Introduction

Consider a flow $\varphi\colon \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ generated by a differential equation $\dot{x} = f(x)$, $f \in C^r(\mathbb{R}^n, \mathbb{R}^n)$.

Definition 15.1.1. Let $\varphi\colon \mathbb{R} \times X \to X$ be a flow and let $U \subset X$. The *alpha* and *omega limit sets* of U are defined by

$$\alpha(U) = \alpha(U, \varphi) \stackrel{\text{def}}{=} \bigcap_{T \le 0} \overline{\varphi\big((-\infty, T], U\big)} \quad \text{and} \quad \omega(U) = \omega(U, \varphi) \stackrel{\text{def}}{=} \bigcap_{T \ge 0} \overline{\varphi\big([T, \infty), U\big)},$$

respectively.

In a slight abuse of notation, if $x \in X$ then we will let $\omega(x) = \omega(\{x\})$ and $\alpha(x) = \alpha(\{x\})$. As an exercise we let the reader check that if x_0 is a heteroclinic point from x_- to x_+, then $\alpha(x_0) = x_-$ and $\omega(x_0) = x_+$.

Given an invariant set $S \subset \mathbb{R}^n$ we can define the associated stable and unstable manifolds by

$$W^s(S) := \{x \in \mathbb{R}^n \mid \omega(x) \subset S\} \quad \text{and} \quad W^u(S) := \{x \in \mathbb{R}^n \mid \alpha(x) \subset S\}.$$

Assume that $S_0, S_1 \subset \mathbb{R}^n$ are invariant sets (it is possible that $S_0 = S_1$) and consider an initial condition x_0 such that $\omega(x) \subset S_1$ and $\alpha(x) \subset S_0$, then by definition $x \in W^s(S_1) \cap W^u(S_0)$. Obviously, the converse is also true, if $x_0 \in W^s(S_1) \cap W^u(S_0)$, then $\omega(x) \subset S_1$ and $\alpha(x) \subset S_0$ and in fact $\varphi(\mathbb{R}, x_0) \subset W^s(S_1) \cap W^u(S_0)$. The point being emphasized is that understanding the dynamics between invariant sets is equivalent to understanding the intersections of their stable and unstable manifolds.

On this level of generality it is impossible to make meaningful comments about particular systems. There are a variety of different approaches that can be taken at this point: one is to make global restrictions about the structure of the invariant sets, e.g. Axiom

A dynamics (see [34]); another is to put restrictions on the types of invariant sets being considered, e.g isolated invariant sets (see [14]). Since we are interested in a constructive approach in this Chapter we restrict our attention to invariant sets that are hyperbolic fixed points. The immediate and fundamental consequence is that we can make use of the Stable and Unstable Manifold Theorem. Recall that if $\widehat{x}$ is a hyperbolic fixed point, then the local stable and unstable manifolds, $W^s_{\rm loc}(\widehat{x})$ and $W^u_{\rm loc}(\widehat{x})$, are C^r manifolds of dimension n_s and n_u, respectively, where $n_s + n_u = n$. Furthermore, there are parameterizations of the local stable and unstable manifolds; that is, C^r functions $P_{\widehat{x}} \colon U \to \mathbb{R}^n$ and $Q_{\widehat{x}} \colon V \to \mathbb{R}^n$, where $U \subset \mathbb{R}^{n_s}$ and $V \subset \mathbb{R}^{n_u}$ are open neighborhoods of the origin, such that

$$P_{\widehat{x}}(U) = W^s_{\rm loc}(\widehat{x}) \quad \text{and} \quad Q_{\widehat{x}}(V) = W^u_{\rm loc}(\widehat{x}).$$

On a more global level this implies that stable and unstable manifolds, $W^s(\widehat{x})$ and $W^u(\widehat{x})$, are immersed manifolds in $\mathbb{R}^n$. As is shown in Example **??** in the setting of a homoclinic orbit $W^s(\widehat{x})$ and $W^u(\widehat{x})$ fail to be embeddings in $\mathbb{R}^n$. The results of Chapter **??** show that this failure can happen more generally and much more spectacularly. Nevertheless, this framework allows us to replace the question of intersection of stable and unstable manifolds by the question of intersection of images of two functions.

It should not be a surprise that actually applying the above mentioned ideas raises technical issues. To make these challenges clear we begin with a definition and a precise statement of our discussion.

Definition 15.1.2. *Consider a differential equation $\dot{x} = f(x)$, $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$, with equilibria $\widehat{x}$ and $\widehat{y}$. A point $x_0 \in \mathbb{R}^n$ is a* heteroclinic point from $\widehat{x}$ to $\widehat{y}$ *if*

$$\alpha(x_0) = \widehat{x} \quad \textit{and} \quad \omega(x_0) = \widehat{y}.$$

If $\widehat{x} = \widehat{y}$, then x_0 is a homoclinic point. *The full solution $\varphi(\mathbb{R}, x_0)$ is called a* heteroclinic orbit from $\widehat{x}$ to $\widehat{y}$ *or* homoclinic orbit to $\widehat{x}$.

The previous discussion can be summarized in the following proposition.

Proposition 15.1.3. *Consider a differential equation $\dot{x} = f(x)$, $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ for which $\widehat{x}$ and $\widehat{y}$ are hyperbolic equilibria. Let $Q_{\widehat{x}} \colon U \subset \mathbb{R}^{n_u} \to \mathbb{R}^n$ and $P_{\widehat{y}} \colon V \subset \mathbb{R}^{n_s} \to \mathbb{R}^n$ denote parameterizations of $W^u_{\rm loc}(\widehat{x})$ and $W^s_{\rm loc}(\widehat{y})$. Then there exists a heteroclinic orbit from $\widehat{x}$ to $\widehat{y}$ if and only if there exists a point $x_0 \in \mathbb{R}^n$ and times $t_- \leq t_+$ such that*

$$\varphi(t_-, x_0) \in Q_{\widehat{x}}(U) \quad \textit{and} \quad \varphi(t_+, x_0) \in P_{\widehat{y}}(V).$$

A special case of this proposition is when there exists an x_0 such that $t_- = t_+$ which leads to the following definition.

Definition 15.1.4. *Given hyperbolic equilibria $\widehat{x}$ and $\widehat{y}$ of $\dot{x} = f(x)$, $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ and parameterizations of $W^u_{\text{loc}}(\widehat{x})$ and $W^s_{\text{loc}}(\widehat{y})$ given by $Q_{\widehat{x}} : U \subset \mathbb{R}^{n_u} \to \mathbb{R}^n$ and $P_{\widehat{y}} : V \subset \mathbb{R}^{n_s} \to \mathbb{R}^n$, a point $x_0 \in \mathbb{R}^n$ is a* short heteroclinic point *if*

$$x_0 \in Q_{\widehat{x}}(U) \cap P_{\widehat{y}}(V).$$

The associated orbit $\varphi(\mathbb{R}, x_0)$ is called a short heteroclinic orbit*. A heteroclinic point which does not lie on a short heteroclinic orbit is called a* long heteroclinic orbit*.*

If one attempts to apply Proposition 15.1.3 to a specific problem, then it becomes immediately clear that we do not have an explicit representation for $Q_{\widehat{x}}$ nor $P_{\widehat{y}}$. However, Theorem **??** provides us with explicit matrices $A_{s,\widehat{y}}$ and $A_{u,\widehat{x}}$, explicit constants ν_s, ν_u and δ_s, δ_u, and the existence of analytic functions $h_{s,\widehat{y}}$ and $h_{u,\widehat{x}}$ such that

$$P_{\widehat{y}}(\theta) = \widehat{y} + A_{s,\widehat{y}}\theta + h_{s,\widehat{y}}(\theta) \quad \text{where} \quad \|h\|_{\nu_s} \leq \delta_s$$

and

$$Q_{\widehat{x}}(\phi) = \widehat{x} + A_{u,\widehat{x}}\phi + h_{u,\widehat{x}}(\phi) \quad \text{where} \quad \|h\|_{\nu_u} \leq \delta_u.$$

Thus we need to extend Proposition 15.1.3 in such a way that the hypotheses are dependent upon this data, e.g. if we are trying to prove the existence of short heteroclinic orbits, then we need to derive conditions under which it is sufficient to prove the existence of a solution to

$$\widehat{y} + A_{s,\widehat{y}}\theta + h_{s,\widehat{y}}(\theta) = \widehat{x} + A_{u,\widehat{x}}\phi + h_{u,\widehat{x}}(\phi)$$

without an explicit expression for $h_{s,\widehat{y}}$ and $h_{u,\widehat{x}}$.

With this in mind we strengthen the notion of intersection of manifolds.

Definition 15.1.5. *Let M_1 and M_2 be C^1 embeddings of disks in $\mathbb{R}^n$, i.e. there exist one-to-one C^1 functions $g_i : \mathbb{R}^{n_i} \to \mathbb{R}^n$ such that $g_i(\mathbb{R}^{n_i}) = M_i$, $i = 1, 2$. M_1 and M_2 are* transverse at $x \in \mathbb{R}^n$ *if either*

1. *$x \notin M_1 \cap M_2$, or*
2. $$Dg_1(g_1^{-1}(x))\mathbb{R}^{n_1} \oplus Dg_2(g_2^{-1}(x))\mathbb{R}^{n_2} = \mathbb{R}^n.$$

This is denoted by $M_1 \pitchfork_x M_2$. M_1 and M_2 are transverse *if they are transverse for all $x \in \mathbb{R}^n$.*

We leave the proof of the following fundamental result as an exercise.

Proposition 15.1.6. *Let $g_i : \mathbb{R}^{n_i} \to \mathbb{R}^n$ be one-to-one C^1 functions such that $g_i(\mathbb{R}^{n_i}) = M_i$, $i = 1, 2$ and M_1 is transverse to M_2 at x. If $\tilde{g}_i$ is sufficiently close to g_i in the C^1 norm, and $\tilde{g}_i(\mathbb{R}^{n_i}) = \tilde{M}_i$, then $\tilde{M}_1$ is transverse to $\tilde{M}_2$ at x.*

Our interest in transverse intersections is that Proposition 15.1.6 suggests that it can be detected even in the presence of numerical error, assuming of course that the numerical error is sufficiently small. The two examples that follow are fundamental to dynamics.

Example 15.1.7. *Let $\widehat{x}$ and $\widehat{y}$ be hyperbolic equilibria of $\dot{x} = f(x)$, $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ with unstable and stable manifolds of dimension n_u and n_s respectively. Assume that x_0 is a heteroclinic point from $\widehat{x}$ to $\widehat{y}$. If $W^u(\widehat{x}) \pitchfork_{x_0} W^s(\widehat{y})$, then the fact that $f(x) \in T_{x_0}W^u(\widehat{x})$ and $f(x) \in T_{x_0}W^s(\widehat{y})$ and a simple dimension count implies that $n_u + (n - n_s) \geq n + 1$, or equivalently*

$$n_u - n_s \geq 1.$$

In particular, if x_0 is a homoclinic point to $\widehat{x}$, then $W^u(\widehat{x})$ and $W^s(\widehat{x})$ cannot intersect transversally at x_0.

Example 15.1.8. *Let Γ be a hyperbolic periodic orbit of $\dot{x} = f(x)$, $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ with Floquet multipliers $\{\sigma_i \mid i = 1, \ldots, n-1\} \cup \{1\}$ such that $|\sigma_i| < 1$ for $i = 1, \ldots, n_s$ and $|\sigma_i| > 1$ for $i = n_s + 1, \ldots, n-1$. Then the dimensions of $W^u(\Gamma)$ and $W^s(\Gamma)$ are $n - n_s$ and $n_s + 1$, respectively. In particular, this implies that it is possible that there exists a point $x_0 \in W^u(\Gamma) \cap W^s(\Gamma)$ such that*

$$W^u(\Gamma) \pitchfork_{x_0} W^s(\Gamma).$$

Examples 15.1.7 and 15.1.8 focus on transverse intersections of stable and unstable manifolds at a single point x_0. However, the transversality extends to the entire orbit $\varphi(\mathbb{R}, x_0)$. We do not provide a proof of this result, but intuition behind it is reasonably straightforward. By definition $W^u \pitchfork_{x_0} W^s$ implies that there exists a basis for $T_{x_0}\mathbb{R}^n$ using vectors from $T_{x_0}W^u$ and $T_{x_0}W^s$. Using these basis vectors construct a matrix $\Psi_0 = [\Psi_0^u | \Psi_0^s]$ where the columns of Ψ_0^u are constructed using vectors from $T_{x_0}W^u$ and the columns of Ψ_0^s are constructed using vectors from $T_{x_0}W^s$. Let $\Psi(t)$ denoted the fundament matrix solution to the IVP $\dot{\Psi} = D(\varphi(t, x_0))\Psi$, $\Psi(0) = \Psi_0$. Using the fact that stable and unstable manifolds are invariant under the flow one can check that if $\Psi(t) = [\Psi(t)^u | \Psi(t)^s]$, then $\Psi(t)^u$ and $\Psi(t)^s$ are made up of vectors from $T_{\varphi(t,x_0)}W^u$ and $T_{\varphi(t,x_0)}W^s$. The following proposition is a formal statement of this result.

Proposition 15.1.9. *Let $\widehat{x}$ and $\widehat{y}$ be hyperbolic equilibria of $\dot{x} = f(x)$, $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$. If $W^u(\widehat{x}) \pitchfork_{x_0} W^s(\widehat{y})$, then $W^u(\widehat{x}) \pitchfork_y W^s(\widehat{y})$ for all $y \in \varphi(\mathbb{R}, x_0)$.*

Similarly let Γ_1 and Γ_2 be periodic orbits of $\dot{x} = f(x)$. If $W^u(\Gamma_1) \pitchfork_{x_0} W^s(\Gamma_2)$, then $W^u(\Gamma_1) \pitchfork_y W^s(\Gamma_2)$ for all $y \in \varphi(\mathbb{R}, x_0)$.

Part IV

Nonlinear Dynamics

Chapter 16

Dynamical Aspects of ODEs

16.1 Conley's Decomposition Theorem

Corresponding results hold for repellers. Attractors and repellers provide a powerful tool for decomposing dynamics. We shall restrict our attention to decomposition of compact invariant sets.

Definition 16.1.1. Let A be an attractor for a flow $\varphi\colon \mathbb{R}\times X \to X$ on a compact metric space. The *dual repeller* to A is

$$A^* \stackrel{\text{def}}{=} \{x \in X \mid \omega(x) \cap A = \emptyset\}.$$

(A, A^*) is called an *attractor-repeller pair decomposition* for φ.

Lemma 16.1.2. *Let (A, A^*) be an attractor repeller pair decomposition, then*

$$A \cap A^* = \emptyset.$$

Proof. Since A is an attractor there exists a trapping region N such that $A = \omega(N) \subset \operatorname{int}(N)$. Observe that this implies that for any $y \in N$, $\omega(y) \subset A$. Hence $A^* \cap N = \emptyset$ and therefore $A \cap A^* \subset \operatorname{int}(N) \cap A^* \subset N \cap A^* = \emptyset$. $\square$

Returning to the logistic equation in Example 5.1.4 and the induced flow $\varphi\colon \mathbb{R}\times[0,K] \to [0,K]$ observe that $\{K\}$ is an attractor. Its dual repeller is $\{0\}$, thus $(\{K\}, \{0\})$ forms an attractor-repeller pair decomposition for φ. Observe that contrary to the name, this is not a decomposition of the phase space $[0,K]$. The justification for calling it a decomposition is made clear in Theorem 16.1.4. It is also worth noting that $\{0\}$ is a repeller. As the following result, which is left as an exercise, indicates this is true in general.

Proposition 16.1.3. *If A is an attractor for a flow $\varphi\colon \mathbb{R}\times X \to X$ on a compact metric space, then its dual repeller A^* is a repeller for φ.*

Theorem 16.1.4. *Let $\varphi\colon \mathbb{R} \times X \to X$ be a flow on a compact metric space. Let (A, A^*) be an attractor-repeller pair decomposition for φ. If $x \in X \setminus (A \cup A^*)$, then*

$$\omega(x) \subset A \quad \textit{and} \quad \alpha(x) \subset A^*.$$

Proof. Since A is an attractor there exists a trapping region N such that $A = \omega(N) \subset \operatorname{int}(N)$. Observe that this implies that for any $y \in N$, $\omega(y) \subset A$.

We want to prove that $\omega(x) \subset A$. By definition if $x \notin A^*$, then $\omega(x) \cap A \neq \emptyset$. Therefore, by Proposition 5.1.10(i) there exists $t > 0$ such that $\varphi(t, x) = y \in N$. By Proposition 5.1.10(ii), $\omega(x) = \omega(y) \subset A$.

We want to prove that $\alpha(x) \subset A^*$. We first show that $\alpha(x) \cap A = \emptyset$. Assume not. Then, by the analogue of Proposition 5.1.10(i) for alpha limit sets, there exists a sequence of times $t_k \to -\infty$ such that

$$\lim_{k\to\infty} \varphi(t_k, x) = \lim_{k\to\infty} y_k = y \in A.$$

Thus without loss of generality we can assume that $y_k \in N$ for all k. Since N is a trapping region $\varphi([0, \infty), y_k) \subset N$. Since this is true for all t_k, we can conclude that $\varphi(\mathbb{R}, x) \subset N$. This implies that x belongs to the maximal invariant set in N and hence, by Proposition 5.1.14 that $x \in A$. Therefore, $\alpha(x) \cap A = \emptyset$.

By the analogue of Proposition 5.1.10(iv) for alpha limit sets, $\alpha(x) \neq \emptyset$. Assume $y \in \alpha(x)$. Since $\alpha(x)$ is invariant, $\omega(y) \subset \alpha(x)$. This implies that $\omega(y) \cap A = \emptyset$ and hence by definition $y \in A^*$. $\square$

Theorem 16.1.5. *Let (A, A^*) be an attractor-repeller pair decomposition for $\varphi\colon \mathbb{R} \times X \to X$, a flow on a compact metric space. Then, there exists a continuous function $V\colon X \to [0, 1]$ such that*

(i) If $t > 0$, then $V(\varphi(t, x)) \leq V(x)$;

(ii) If $t > 0$ and $V(\varphi(t, x)) = V(x)$, then $x \in A \cup A^*$;

(iii) $V(x) = 1$ if and only if $x \in A^*$, and $V(x) = 0$ if and only if $x \in A$.

Proof. $\square$

Attractor-repeller pair decompositions for the dynamics of Examples 5.1.4 and 5.1.8 are reasonably easy to describe: $(\{K\}, \{0\})$ and $\left(\left\{x \in \mathbb{R}^2 \;\middle|\; \|x\| = |K|\right\}, \{0\}\right)$, respectively. A significant feature distinguishing the systems is that in the first case the attractor is an equilibrium, while in the second case it is a periodic orbit. It should also be observed that both systems depend on two parameters, r and K for the logistics equation and λ and K for Examples 5.1.8, and that as one changes these parameters the solutions to the differential equations change. This is made explicit by (5.2). This raises the question of what level of refinement do we want to use to distinguish between the dynamics of different systems of differential equations.

16.2 Exercises

Exercise 16.2.1. Prove Proposition 5.1.6.

Exercise 16.2.2. Prove that given a flow $\varphi\colon \mathbb{R} \times X \to X$ and a set $A \subset X$

$$\bigcup_{x \in A} \omega(x, \varphi) \subset \omega(A, \varphi).$$

Provide an example for which

$$\bigcup_{x \in A} \omega(x, \varphi) \neq \omega(A, \varphi).$$

Exercise 16.2.3. Prove Proposition 16.1.3.

Exercise 16.2.4. Let $f\colon \mathbb{R}^n \to \mathbb{R}^n$ be C^1. Show that the flows generated by

$$\dot{x} = f(x) \quad \text{and} \quad \dot{x} = \frac{f(x)}{1 + \|f(x)\|}$$

are topologically equivalent.

Exercise 16.2.5. Observe that the definition of topologically equivalent in Definition 4.2.2 requires a homeomorphism $h\colon X \to Y$ as opposed to diffeomorphism. Since every diffeomorphism is a homeomorphism, our definition leads to larger equivalence classes. Using scalar differential equations provide an explanation for this choice. More explicitly, consider the family of linear scalar differential equation $\{\dot{x} = ax \mid a \in \mathbb{R}\}$.

(i) Determine the topological equivalence classes.

(ii) In the definition of topological equivalence replace the homeomorphism $h\colon X \to Y$ with the requirement that h be a diffeomorphism. Determine the equivalence classes.

The point of this exercise is to show that if we use a diffeomorphism, then the equivalence relation requires that the eigenvalues of the derivative at the equilibrium be the same.

Exercise 16.2.6. Show that if x_0 is a heteroclinic point from x_- to x_+, then $\alpha(x_0) = x_-$ and $\omega(x_0) = x_+$.

Exercise 16.2.7. Give an example of a flow $\varphi\colon \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2$ for which there exists a point $x \in \mathbb{R}^2$ such that $\omega(x) \neq \emptyset$, but $\omega(x)$ is not connected.

Exercise 16.2.8 (Lyapunov's stability theorem). Let $\tilde{x}$ be an equilibrium for $\dot{x} = f(x)$, $x \in \mathbb{R}^n$ where $f \in C^1(U, \mathbb{R}^n)$ for some open set $U \subset \mathbb{R}^n$. Let W be a neighborhood of $\tilde{x}$, $V\colon W \to \mathbb{R}$ be a continuous function that is differentiable on $W \setminus \{\tilde{x}\}$. Define

$$\dot{V}(x) \stackrel{\text{def}}{=} \frac{d}{dt} V(x(t)) = DV(x) \cdot \dot{x}(t) = DV(x) \cdot f(x),$$

and assume that V satisfies the following properties:

(i) $V(\tilde{x}) = 0$ and $V(x) > 0$ if $x \in W \setminus \{\tilde{x}\}$.

(ii) $\dot{V}(x) \leq 0$ for all $x \in W \setminus \{\tilde{x}\}$.

Then, $\tilde{x}$ is stable. Furthermore, if in addition $\dot{V}(x) < 0$ for all $x \in W \setminus \{\tilde{x}\}$, then $\tilde{x}$ is asymptotically stable.

Chapter 17

Chaotic Dynamics

17.1 Introduction

Example 15.1.8 suggests that it is possible for the stable and unstable manifolds of a hyperbolic periodic orbit to intersect transversally. The goal of this chapter is to demonstrate that the occurrence of this phenomena leads to an extremely complicated invariant set that is one of the archetypical examples of what is commonly referred to as chaotic dynamics.

We begin our discussion by considering one of the fundamental examples of chaotic dynamics the Lorenz equations (2.27). Figure 17.1 shows numerical images of the first return map for some points in the rectangle $[-8,8]\times[-2.5,2.5]\times\{53\}\subset\mathbb{R}^3$ for the Lorenz equations with parameter values $(\sigma,\beta,\rho) = (45,10,54)$. There are set of initial values aligned on the boundary of two parallelograms N_0 and N_1. Using the tools developed in Chapter 10 the reader can prove that in each parallelogram there exists a hyperbolic periodic orbit Γ_i such that $\Gamma_i \cap N_i = \gamma_i \in \mathbb{R}^3$, $i = 0,1$ and $\Gamma_i \cap N_j = \emptyset$ for $i \neq j$. Using the tools developed in Chapter 8 and in Chapter 9 the reader can prove that the stable manifold for the equilibrium point $(0,0,0)$ intersects the rectangle separating the two parallelograms. In particular for these points there is no first return time and thus the Poincaré map cannot be extended over the entire rectangle. Using the tools developed in Chapter 11 the reader can prove that the Floquet multipliers $\{\sigma_1,\sigma_2\}\cup\{1\}$ for the periodic orbits have the property that $|\sigma_1|<1$ and $|\sigma_2|>1$. In a future version of these notes tools will be provided that allow the reader to compute $W^s_{\text{loc}}(\Gamma_i)$ and $W^u_{\text{loc}}(\Gamma_j)$ the local stable and unstable manifolds for these periodic orbits and demonstrate that the global stable and unstable manifolds intersect transversally, i.e. $W^u(\Gamma_i) \pitchfork W^s(\Gamma_j)$.

However, as stated above, the goal of this chapter is to explain the consequences of these observations. Thus we reduce this information to the idealized images presented in Figure 17.2 and in Figure 17.3. On the region $R = N_0 \cup N_1$ the Poincaré map P is defined. The images of $P(N_0)$ and $P(N_1)$ are indicated via the shaded regions. The intersection of the periodic orbits Γ_i with R is indicated by the point γ_i, respectively. To

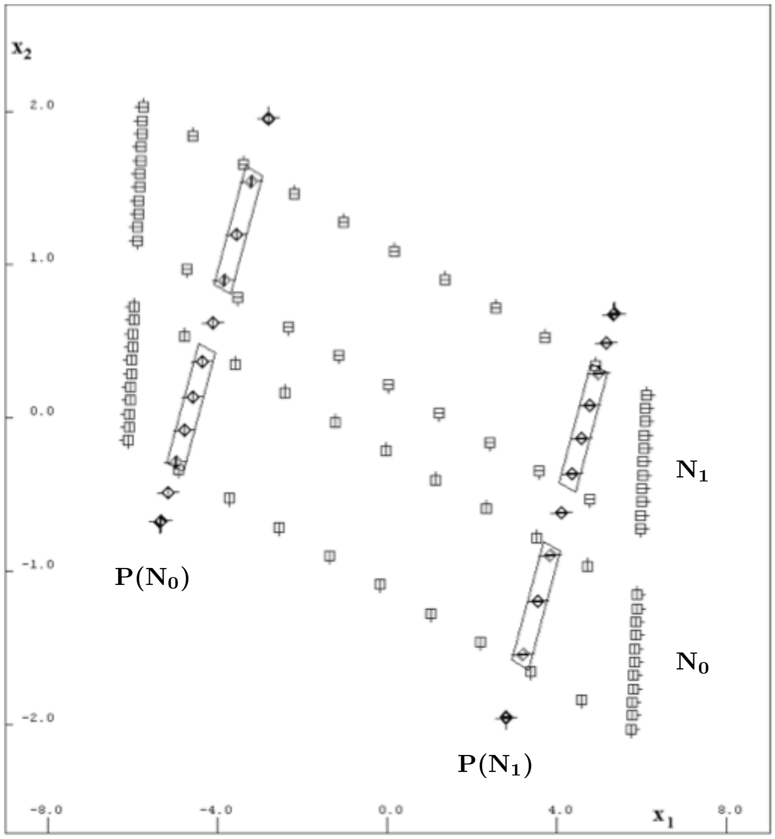

Figure 17.1: Poincaré section and first return images for the Lorenz equations at parameter values $(\sigma, \beta, \rho) = (45, 10, 54)$. The Poincaré section is contained in the rectangle $[-8, 8] \times [-2.5, 2.5] \times \{53\} \subset \mathbb{R}^3$. The square boxes indicate the locations of the initial values and the diamonds indicate the first return values. The initial values are taken from the boundary of two parallelograms N_0 and N_1. The return values for these points, denoted here by $P(N_0)$ and $P(N_1)$ appear to be lines, however, this is due to the fact that the flow of Lorenz has a strong contraction in one direction.

avoid any unnecessary complications we have drawn the intersection of the local stable and unstable manifolds of Γ_1 and Γ_2 as horizontal and vertical lines, clearly indicating a transverse intersection. In analogy with Proposition 15.1.3 it is clear that any point on the intersection of these stable and unstable manifolds are connecting points between the periodic orbits.

Observe that $P(R) \not\subset R$ and thus there are points $x_0 \subset R$ such that $P^n(x_0) \notin R$ for some $n \in \mathbb{N}$. Similarly, since $R \not\subset P(R)$ there are points $x_0 \subset R$ such that $P^{-n}(x_0) \notin R$ for some $n \in \mathbb{N}$. In Section 17.3 we provide a complete characterization of the dynamics

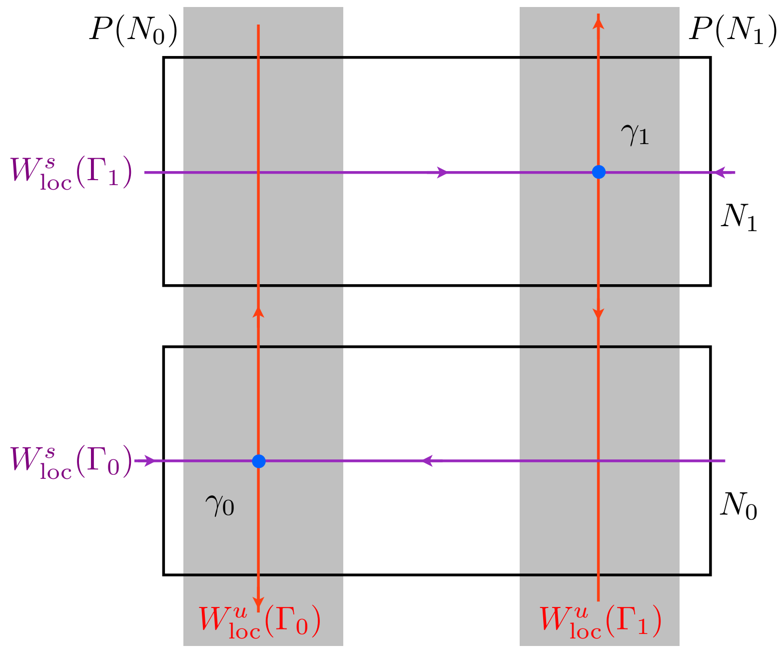

Figure 17.2: Idealized return map for the Lorenz equations at parameter values $(\sigma, \beta, \rho) = (45, 10, 54)$. The intersection of Γ_i with the Poincaré section is indicated by the point γ_i. The horizontal purple lines indicate the intersection of the local stable manifolds of Γ_i with the Poincaré section and the vertical red lines indicate the intersection of the local unstable manifolds of Γ_i with the Poincaré section. The points of intersection between the red and purple lines are connecting points between the periodic orbits. The image of N_i under P is indicated by the shaded regions.

of points $x \in R$ such that $P^n(x) \in R$ for all $n \in \mathbb{Z}$.

17.2 Symbolic Dynamics

As is demonstrated in the next section, the dynamics of the Lorenz equations is extremely complicated and thus we need a language that simply and explicitly expresses this complication. To do this we introduce the concept of symbolic dynamics. The key feature is that the action induced by the map is extremely simple, but it is defined on a complicated space. Thus the complexity of the dynamics is associated with the points in the space as opposed to the map itself.

Definition 17.2.1. The *symbol space* on n symbols is defined to be

$$\Sigma_n := \{1, \dots, n\}^{\mathbb{Z}}$$

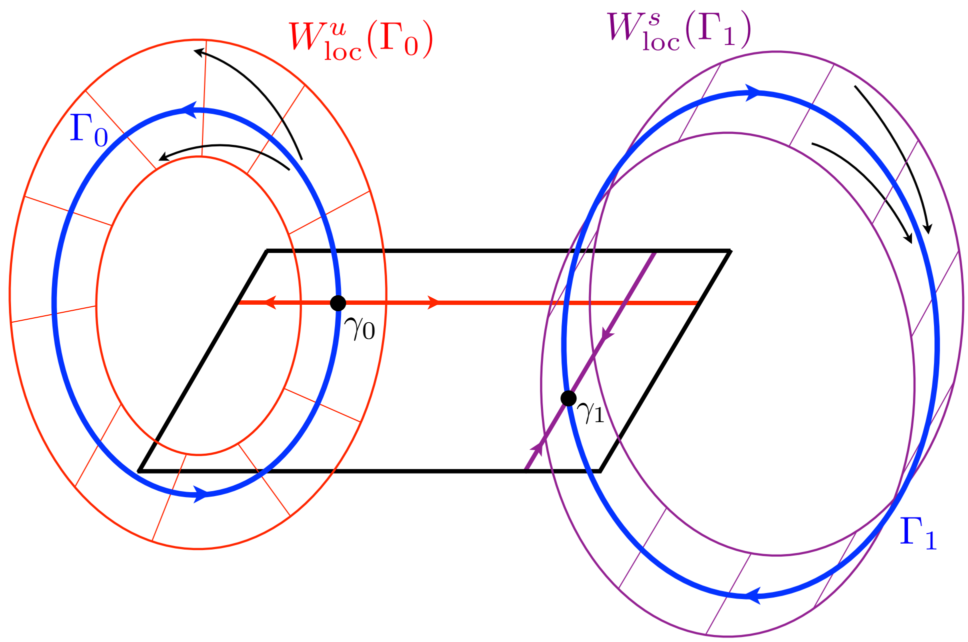

Figure 17.3: Visualization of $W^u(\Gamma_0)$ and $W^s(\Gamma_1)$ in the case of an idealized return map for the Lorenz equations at parameter values $(\sigma, b, r) = (45, 10, 54)$. The intersection of Γ_i with the Poincaré section is indicated by the point γ_i. The thick purple line indicates the intersection of $W^s_{\mathrm{loc}}(\Gamma_1)$ with the Poincaré section and the thick red line indicates the intersection of $W^u_{\mathrm{loc}}(\Gamma_0)$ with the Poincaré section. It is not hard to imagine that $W^u(\Gamma_0) \pitchfork W^s(\Gamma_1)$.

with the metric

$$d(a,b) := \sum_{k\in\mathbb{Z}} \delta(a_k, b_k) 4^{-|k|}$$

where

$$\delta(x,y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise.} \end{cases}$$

We leave it to the reader to check that Σ_n is a compact metric space.

Definition 17.2.2. The *shift map* on n symbols, $\sigma\colon \Sigma_n \to \Sigma_n$, is defined by

$$(\sigma(a))_k = a_{k+1}.$$

We leave it to the reader to check that σ is a uniformly continuous map. Observe that any periodic sequence in Σ_n corresponds to a periodic orbit under σ. Furthermore, there exists infinitely many connecting orbits between any two periodic orbits and uncountably many orbits whose ω limit set is the entire space Σ_n

In the context of ODEs, symbolic dynamics is typically used to obtain lower bounds on the complexity of the dynamics. This is done by constructing a conjugacy or semi-conjugacy between dynamics on a Poincaré section of the flow with invariant subset of the dynamics of $\sigma\colon \Sigma_n \to \Sigma_n$.

Definition 17.2.3. A *transition matrix* on n symbols is an $n \times n$ matrix with $0,1$ entries for which each row and each column has at least one nonzero entry.

Definition 17.2.4. Let $A = (a_{j,k})$ be a transition matrix on n symbols. The associated *subshift space* is given by

$$\Sigma_A := \left\{ s \in \Sigma_n \mid a_{s_k, s_{k+1}} = 1,\ k \in \mathbb{Z} \right\}.$$

The associated subshift dynamics is denoted by $\sigma_A := \sigma|_{\Sigma_A}$ and obtained by restricting σ to Σ_A.

Proposition 17.2.5. *If A is a transition matrix on n symbols, then Σ_A is a compact invariant subset of Σ_n.*

17.3 Smale Horseshoe

The discussion in Section 17.1 is meant to motivate the following classic example of chaotic dynamics known as the Smale horseshoe. This is a map $P\colon R \to R$ where $R \subset \mathbb{R}^2$ is a rectangle. We use Figure 17.4 to provide a geometric as opposed to analytic description of P. To understand the action of P it is best to consider its construction in three consecutive steps:

1. the rectangle R is contracted uniformly in the horizontal direction and expanded uniformly in the vertical direction, i.e. if R has width a and height b, then then after the first step we have a new rectangle with width λa and height μb where $0 < \lambda < \frac{1}{2}$ and $2 < \mu$;

2. the new rectangle is bent into the shape of a horseshoe; and

3. the horseshoe is laid over the original rectangle R as indicated in Figure 17.4.

Under this construction P can be extended to a diffeomorphism on all of $\mathbb{R}^2$ and we assume that this is done. The exact extension is not of interest to us as we focus on describing the dynamics of

$$S := \{x \in N_0 \cup N_1 \mid P^n(x) \in N_0 \cup N_1, n \in \mathbb{Z}\} = \bigcap_{n \in \mathbb{Z}} P^n(N_0 \cup N_1).$$

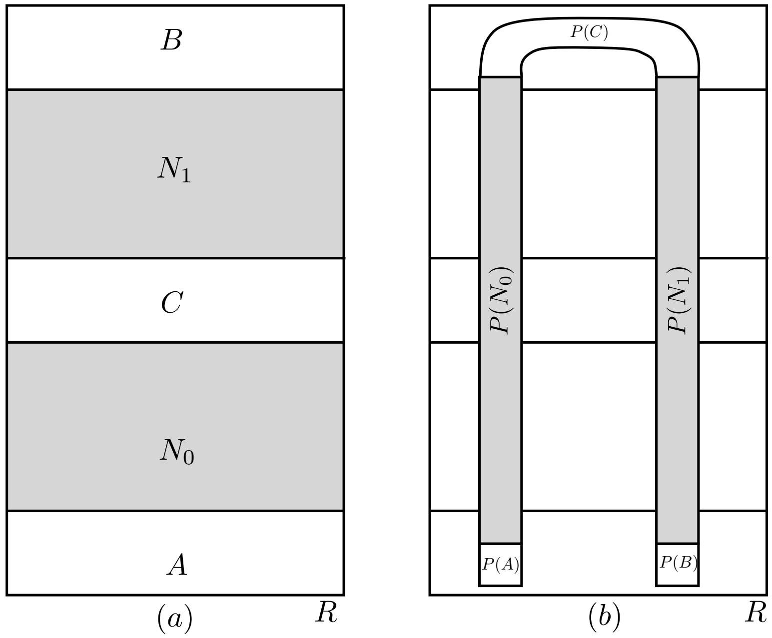

Figure 17.4: The Smale horseshoe.

A trivial observation, but one that sets the stage for how the dynamics on S can be described, is that if $x \in S$, then

$$P^{-1}(x) \in N_0 \cup N_1, \quad x = P^0(x) \in N_0 \cup N_1 \quad \text{and} \quad P(x) \in N_0 \cup N_1.$$

Thus, a simple way of keeping track of the location of elements of S under the action of P is through an itinerancy map $\rho \colon S \to \{0,1\}^{\mathbb{Z}}$ defined by

$$\rho(x)_n = \begin{cases} 0 & \text{if } P^n(x) \in N_0, \\ 1 & \text{if } P^n(x) \in N_1. \end{cases}$$

By Tychonoff's Theorem [26] applying the product topology to $\Sigma := \{0,1\}^{\mathbb{Z}}$ implies that it is a compact space and the fact that N_0 and N_1 are disjoint compact sets can be used to prove that ρ is a continuous function.

Let $\sigma \colon \Sigma_2 \to \Sigma_2$ be the shift map, that is given $s = (\ldots, s_{-1}, s_0, s_1, \ldots) \in \Sigma$ define

$$\sigma(s)_n := s_{n+1}.$$

Observe that this gives rise to the following relationship between σ and P,

$$\rho(P(x)) = \sigma(\rho(x)) \tag{17.1}$$

for all $x \in S$.

Theorem 17.3.1. *Given the map P defined above, the itinerancy map $\rho\colon S \to \Sigma$ is a homeomorphism and hence provides a topological conjugacy between the dynamics of P acting on S and σ acting on Σ.*

Before turning to the proof of this theorem it is worth commenting on its implications. In particular, assume that the action of P on N_0 and N_1 arises as the Poincaré map for a differential equation $\dot{x} = f(x)$, $x \in \mathbb{R}^3$ with flow φ.

1. Since $\rho\colon S \to \Sigma$ is a homeomorphism it is a bijection. Since $\Sigma = \{0,1\}^{\mathbb{Z}}$ is an uncountable set, S contains uncountably many points. In particular, S contains uncountably many limit points.

2. Assume the height of C in Figure 17.4 is κ. Assume $x, x' \in S$ are distinct points. Let $s = \rho(x)$ and $s' = \rho(x')$. Since ρ is a bijection, $s \neq s'$. Thus there exists $n \in \mathbb{Z}$ such that $s_n \neq s'_n$. Without loss of generality assume $s_n = 0$ and $s'_n = 1$. Then $P^n(x_n) \in N_0$ and $P^n(x'_n) \in N_1$. In the context of the solutions to the differential equation this implies that there exists $T \in \mathbb{R}$ such that $\|\varphi(T,x) - \varphi(T,x')\| > \kappa$.

   This observation is true for any two distinct elements of S. If $x \in S$ is a limit point, then there exists $\{x_k \in S\}$ such that $\lim_{k\to\infty} x_k = x$. This implies that there is a sequence of times T_k such that

   $$\|\varphi(T_k, x) - \varphi(T_k, x_k)\| > \kappa.$$

   Because the flow φ is continuous we know that $\lim_{k\to\infty} |T_k| = \infty$, but nevertheless we can claim that arbitrarily small perturbations in initial conditions leads to observably distinct trajectories. This is typically called *sensitivity with respect to initial conditions* and is a hallmark of chaotic dynamics.

3. The action of $\sigma\colon \Sigma \to \Sigma$ can be used to characterize the orbits in S. For example, it is immediate that there are exactly two fixed points of P, x_0 and x_1 where

   $$\rho(x_0) = (\ldots, 0, 0, 0, 0, \ldots) \in \Sigma, \quad \text{and} \quad \rho(x_1) = (\ldots, 1, 1, 1, 1, \ldots).$$

   Similarly, any $s \in \Sigma$ represented by a periodic sequence of 0s and 1s corresponds to a unique periodic point of P. Thus, for every period there exists a periodic point, in fact, by enumerating all possible periodic sequences of 0s and 1s on can check that the number of periodic points of period T grows exponentially as a function of T.

   Now consider periodic points x and x' elements of S. These give rise to periodic orbits Γ and Γ' of the differential equation. As above let $s = \rho(x)$ and $s' = \rho(x')$ and define $c \in \Sigma$ by

   $$c = (\ldots, s_{-j-2}, s_{-j-1}, s_{-j}, c_{-j+1}, c_{-j+2}, \ldots, c_{k-2}, c_{k-1}, s'_k, s'_{k+1}, s'_{k+2}, \ldots)$$

where $j, k \in \mathbb{N}$ and $c_i \in [0,1]$ for $i = -j+1, \ldots, k-1$. Observe that $\varphi(\mathbb{R}, \rho^{-1}(c))$ is a connecting orbit from Γ to Γ'. Since j, k and the c_i are arbitrary, this implies that between any two periodic orbits there are infinitely many connecting orbits and similarly there are infinitely many homoclinic orbits to any given periodic orbit.

Finally, observe that each of the orbits described above can be associate with a pair of rational numbers in the interval $[0,1]$, i.e. given s we can associate it with the binary expansions $0.s_0s_1s_2\ldots$ and $0.s_{-1}s_{-2}s_{-3}\ldots$. However, as is indicated in the first comment there are uncountably many distinct orbits and thus the 'typical' orbit in S never limits to a periodic orbit and in this sense exhibits completely unpredictable future and past trajectories.

Proof of Theorem 17.3.1. Given that ρ is continuous it is sufficient to show that ρ is 1-1 and onto.

Assume that $x, x' \in S$, $\rho(x) = \rho(x')$ and $x \neq x'$. Since $S \subset \mathbb{R}^2$ we can write $x = (x_1, x_2)$ and $x' = (x_1', x_2')$. Since $x \neq x'$, $x_1 \neq x_1'$ or $x_2 \neq x_2'$. Assume $x_2 \neq x_2'$. Let ν be the maximum value of the height of N_0 or N_1. Since $\rho(x) = \rho(x')$, $|x_2 - x_2'| \leq \nu$. Observe that there is a first positive integer n such that $\mu^n |x_2 - x_2'| > \nu$. Without loss of generality assume $x_2 > x_2'$. By assumption $\rho(x)_{n-1} = \rho(x')_{n-1}$. If $\rho(x)_{n-1} = \rho(x')_{n-1} = 0$, then $\rho(x)_n = 1$ and $\rho(x')_n = 0$, a contradiction. If $\rho(x)_{n-1} = \rho(x')_{n-1} = 1$, then $\rho(x)_n = 0$ and $\rho(x')_n = 1$, again a contradiction. A similar argument, but letting n be a negative integer applies to the case when $x_1 \neq x_1'$.

To prove that P is onto let $V_i := P(N_i)$. Observe that according to step 1 of the definition of P this is a rectangle of width λa. Let $s \in \Sigma_2$. Observe that

$$P(N_{s_{-1}}) \cap P^2(N_{s_{-2}}) = P(N_{s_{-1}}) \cap P(P(N_{s_{-2}})).$$

Since $P(N_{s_{-2}})$ has width λa, $P(P(N_{s_{-2}}))$ has width $\lambda^2 a$, and hence $P(N_{s_{-1}}) \cap P^2(N_{s_{-2}})$ has width $\lambda^2 a$. Observe that the height of $P(N_{s_{-1}}) \cap P^2(N_{s_{-2}})$ is the same as the height of V_i. By induction

$$\bigcap_{k=1}^{n} P^k(N_{s_{-k}})$$

is a rectangle with width $\lambda^n a$ and height of V_i. Thus

$$\bigcap_{k=1}^{\infty} P^k(N_{s_{-k}})$$

is a vertical line segment. A similar argument shows that

$$\bigcap_{k=-\infty}^{1} P^k(N_{s_{-k}})$$

is a horizontal line segment. Finally, the intersection of these two line segments is a point x such that $\rho(x) = s$. $\square$

We finish this section with the statement of the following fundamental theorem which demonstrates that if an ode contains a hyperbolic periodic orbit then potentially it can exhibit complicated dynamics of the type exhibited by the Smale horseshoe. For a more complete statement and proof the reader is referred to [34, Theorem 4.5].

Theorem 17.3.2. *Assume Γ is a hyperbolic periodic orbit for $\dot{x} = f(x)$, $x \in \mathbb{R}^n$, $n \geq 3$. Assume there exists $x_0 \in \mathbb{R}^n$ such that $x_0 \in W^u(\Gamma) \cap W^s(\Gamma)$ and $W^u(\Gamma) \pitchfork_{x_0} W^s(\Gamma)$. Let $P\colon A \to \Sigma$ be a Poincare map for Γ. Let γ_0 denote the periodic point of Γ that is contained in the domain of P. Assume x_0 is in the domain of P. Then for any neighborhood $B \subset A$ such that $\gamma_0, x_0 \in B$, there exists a positive integer n and an invariant set $S \subset B$ such that the dynamics of P^n restricted to S is conjugate to the shift dynamics on Σ_2.*

Chapter 18

Automatic Differentiation

18.1 Automatic differentiation for power series of one variable

At first glance the example of the previous section suggests that Taylor integrators are practical only for problems with polynomial nonlinearities. In this section we discuss some techniques which let us reduce many problems with analytic nonlinearities to an equivalent polynomial problem. The idea is to exploit that the elementary functions of a standard calculus course are themselves solutions of some simple differential equations.

Consider a power series of one variable

$$P(t) = \sum_{n=0}^{\infty} p_n (t - t_0)^n,$$

and assume (for example) that we want to exponentiate P as a power series, i.e. we want to find a power series

$$Q(t) = \sum_{n=0}^{\infty} q_n (t - t_0)^n,$$

with

$$Q(t) = e^{P(t)}.$$

The idea is that differentiation turns composition of scalar functions into multiplication of scalar functions via the chain rule; and that the exponential function itself solves a simple, linear, differential equation. More precisely, differentiate $Q(t)$ with respect to t gives

$$Q'(t) = e^{P(t)} P'(t) = Q(t) P'(t),$$

or

$$\sum_{n=0}^{\infty}(n+1)q_{n+1}(t-t_0)^n = \left(\sum_{n=0}^{\infty} q_n(t-t_0)^n\right)\left(\sum_{n=0}^{\infty}(n+1)p_{n+1}(t-t_0)^n\right)$$
$$= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n}(n-k+1)p_{n-k+1}q_k\right)(t-t_0)^n,$$

on the level of power series. Matching like powers of $t-t_0$ leads to

$$(n+1)q_{n+1} = \sum_{k=0}^{n}(n-k+1)p_{n-k+1}q_k,$$

and by reindexing and isolating q_n we have

$$q_n = \frac{1}{n}\sum_{k=0}^{n-1}(n-k)p_{n-k}q_k.$$

The technique facilitates efficient computation of power series coefficients for compositions with all the elementary functions. For example if

$$Q(t) = P(t)^{\alpha},$$

with $\alpha \in \mathbb{C}$, then by the chain rule

$$Q'(t) = \alpha P(t)^{\alpha-1}P'(t),$$

or

$$P(t)Q'(t) = \alpha Q(t)P'(t).$$

Computing Cauchy products and matching like powers leads to

$$q_n = \frac{1}{np_0}\sum_{k=0}^{n-1}(n\alpha - k(\alpha+1))p_{n-k}q_k.$$

More examples are considered in the exercises.

Consider now the differential equation

$$x' = -x + xe^{-x}, \qquad \text{with } x(0) = x_0 \in \mathbb{R}.$$

which is a small modification of the example considered in the previous section. We look for a solution of the form

$$x(t) = \sum_{n=0}^{\infty} a_n t^n,$$

with

$$Q(t) = e^{-x(t)} = \sum_{n=0}^{\infty} q_n t^n,$$

where $q_0 = e^{-x_0}$. Plugging the power series into the differential equation leads to

$$\begin{aligned}\sum_{n=0}(n+1)a_{n+1}t^n &= -\sum_{n=0}^{\infty} a_n t^n + \left(\sum_{n=0}^{\infty} a_n t^n\right)\left(\sum_{n=0}^{\infty} q_n t^n\right) \\ &= -\sum_{n=0}^{\infty}\left(a_n + \sum_{k=0}^{n} a_{n-k}q_k\right)t^n.\end{aligned}$$

Matching like powers of t and reindexing leads to the recursion relations

$$a_n = \begin{cases} x_0 & \text{if } n = 0 \\ \frac{1}{n}\left(a_{n-1} + \sum_{k=0}^{n-1} a_{n-k-1}q_k\right) & \text{if } n \geq 1 \end{cases},$$

$$\begin{aligned} q_n &= \begin{cases} e^{-x_0} & \text{if } n = 0 \\ \frac{-1}{n}\sum_{k=0}^{n-1}(n-k)a_{n-k}q_k \end{cases} \\ &= \begin{cases} e^{-x_0} & \text{if } n = 0 \\ -a_n q_0 + \frac{-1}{n}\sum_{k=1}^{n-1}(n-k)a_{n-k}q_k \end{cases}\end{aligned}$$

Remark 18.1.1 (Numerical implementation). Note that q_n depends explicitly on a_n, while a_n depends only on lower order terms. Then when we implement a numerical algorithm based on this recursion scheme we compute first a_n and then have the data necessary to compute q_n.

The power of such "automatic differentiation" schemes is that they reduce transcendental nonlinearities to polynomial. Indeed in the problem just discussed the exponential nonlinearity is evaluated at the same cost (in addition and multiplication operations) as a Cauchy product. The cost is that we must store the additional variables $\{q_0, \ldots, q_n\}$. For many applications, where we compute only a few hundred Taylor coefficients, the cost in additional memory is negligible.

18.1.1 Example: a non-polynomial nonlinearity

Consideration of the recursion scheme for the differential equation

$$x' = -x + xe^{-x},$$

studied in Section 18.1 leads to the fixed point operator

$$\Psi(a, q) = \begin{pmatrix} \Psi_1(a, q) \\ \Psi_2(a, q) \end{pmatrix}$$

where $\Psi_{1,2}\colon \left(\ell^1_{\nu,\mathbb{N}}\right)^2 \to \ell^1_{\nu,\mathbb{N}}$ are the maps defined by

$$\Psi_1(a,q)_n = \begin{cases} x_0 & \text{if } n = 0 \\ \frac{1}{n}\left(a_{n-1} + \sum_{k=0}^{n-1} a_{n-k-1}q_k\right) & \text{if } n \geq 1 \end{cases},$$

$$\Psi_2(a,q)_n = \begin{cases} e^{-x_0} & \text{if } n = 0 \\ -q_0 a_n + \frac{-1}{n}\sum_{k=1}^{n-1}(n-k)a_{n-k}q_k \end{cases}.$$

While the first component operator is exactly as in the previous examples, the operator Ψ_2 is something different. The main problem is that the operator cannot be written as a composition with the operator L^N and S in the same way as in earlier problems, and it is not clear that Ψ_2 is a contraction in its tail.

One way around this problem is to rewrite Ψ_2 as

$$\Psi_2(a,q)_n = \begin{cases} e^{-x_0} & \text{if } n = 0 \\ -q_0\Psi_1(a,q)_n + \frac{-1}{n}\sum_{k=1}^{n-1}(n-k)a_{n-k}q_k \end{cases}.$$

Since Ψ_1 is a contraction in its tail, we can manage the first term. For the second term we obtain an estimate of the form

$$\sum_{n=0}^{\infty}\left|\frac{-1}{n}\sum_{k=1}^{n-1}(n-k)a_{n-k}q_k\right|\nu^n \leq \|a\|_{1,\nu,\mathbb{N}}\|q\|_{1,\nu,\mathbb{N}}\nu.$$

This bound employees the estimate

$$\frac{(n-k)}{n} \leq 1,$$

for all $1 \leq k \leq n-1$ as well as the Banach algebra and the fact that the sum over k does not involve terms of order zero or n (hence we obtain a factor of ν). Since this estimate can be controlled by taking ν small, we have hope of showing that Ψ is a contraction in its tail.

There is however another approach which is perhaps more straight forward. The idea is to go back to the original equation, and use the automatic differentiation on the level of the differential equations, rather than as a recursion formula. So, we introduce the variable

$$y(t) = e^{-x(t)},$$

and note that

$$y'(t) = -x'(t)e^{-x(t)} = -x'(t)y(t).$$

Now, instead of solving this equation via recursion, we use the fact that $x(t)$ solves the differential equation $x' = -x + xe^{-x}$ to obtain

$$y'(t) = -(-x(t) + x(t)e^{-x(t)}))y(t) = x(t)y(t) - x(t)y^2(t).$$

Letting

$$x(t) = \sum_{n=0}^{\infty} t^n,$$

and

$$y(t) = \sum_{n=0}^{\infty} q_n t^n,$$

we consider the system of equations

$$\begin{aligned} x' &= -x + xy \\ y' &= xy - xy^2 \end{aligned}$$

and obtain the recursion relations

$$a_n = \begin{cases} x_0 & \text{if } n = 0 \\ \frac{1}{n}\left(-a_{n-1} + (a * q)_{n-1}\right) & \text{if } n \geq 1 \end{cases} \tag{18.1}$$

and

$$q_n = \begin{cases} e^{-x_0} & \text{if } n = 0 \\ \frac{1}{n}\left((a * q)_{n-1} - (a * q * q)_{n-1}\right) & \text{if } n \geq 1 \end{cases}. \tag{18.2}$$

Proceeding as before we are led to look for fixed points of the operator $T\colon (X^\infty)^2 \to (X^\infty)^2$ defined by

$$T(u,v) = \begin{pmatrix} L^N S\left(-\bar{a} - u + (\bar{a} + u) * (\bar{q} + v)\right) \\ L^N S\left((\bar{a} + u) * (\bar{q} + v) - (\bar{a} + u) * (\bar{q} + v) * (\bar{q} + v)\right) \end{pmatrix},$$

where $\bar{a}, \bar{q} \in X^N$ are obtained by solving the recursion to order N. For this operator we easily obtain the following Lemma, arguing just as in Section **??** (except that we must be careful with the cubic terms). Observe that in this formulation of the error analysis we clearly see the factor of $\nu/(N+1)$ in the derivative bounds, hence have good reason to hope for a contraction when N is large enough. We leave the proof of the Lemma as an exercise.

Lemma 18.1.2. *Define the positive constants*

$$Y_1 \stackrel{\text{def}}{=} \frac{|\bar{a}_N| \nu^{N+1}}{N+1} + \nu \sum_{n=N}^{2N} \frac{1}{n+1} \left| \sum_{k=0}^{n} \bar{a}_{n-k} \bar{q}_k \right| \nu^n,$$

$$Y_2 = \nu \sum_{n=N}^{2N} \frac{1}{n+1} \left| \sum_{k=0}^{n} \bar{a}_{n-k} \bar{q}_k \right| \nu^n, + \nu \sum_{n=N}^{3N} \frac{1}{n+1} \left| \sum_{k=0}^{n} \sum_{j=0}^{k} \bar{a}_{n-k} \bar{q}_{k-j} \bar{q}_j \right| \nu^n,$$

$$Z_{01} \stackrel{\text{def}}{=} 1 + \|\bar{q}\|_N + \|\bar{a}\|_N,$$

$$Z_{02} \stackrel{\text{def}}{=} \|\bar{a}\|_N + \|\bar{q}\|_N + 2\|\bar{a}\|_N\|\bar{q}\|_N + \|\bar{q}\|^2,$$

$$Z_1 \stackrel{\text{def}}{=} 2 + 2\|\bar{a}\|_N + 4\|\bar{q}\|_N.$$

Let

$$Y \stackrel{\text{def}}{=} \max(Y_1, Y_2),$$

and define the polynomial

$$Z(r) \stackrel{\text{def}}{=} \frac{\nu}{N+1}\left(\max(Z_{01}, Z_{02}) + Z_1 r + 3r^2\right).$$

Then

$$\|T(0)\|_{(X^\infty)^2} \leq Y,$$

and

$$\sup_{(u,v)\in\overline{B_r(0)}} \|DT(u,v)\|_{B((X^\infty)^2)} \leq Z(r).$$

Remark 18.1.3 (Numerical implementation)*.* Note that the coefficients $\{\bar{q}_0, \ldots, \bar{q}_N\}$ defined by the recursion in Section 18.1 are the same as defined by the recursion in Equation (18.2) (this is the uniqueness of solutions of ODEs combined with the uniqueness if the Taylor coefficients, i.e. these recursions define the same analytic function). But the recursion in Section 18.1 is only quadratic, hence numerically cheaper to compute. So, we are free to compute the coefficients using one scheme and to validate them using another. We leave the numerical implementation of the coefficient computation and the validation as an exercise.

18.2 The radial gradient and automatic differentiation for multivariate power series

The formal procedure for computing Taylor expansions of invariant manifolds discussed in the previous section is more general than it might at first appear. Recall that automatic differentiation is a tool which let us compute the Taylor series coefficient of compositions of analytic functions for power series of one variable. In two or more variables something similar is possible, but the computations are aided by the introduction of the *radial gradient.*

Definition 18.2.1 (Radial gradient). Suppose that $f\colon \mathbb{R}^m \to \mathbb{R}$. The radial gradient of f is the quantity defined by

$$\langle \nabla f(x), x\rangle = x_1 \frac{\partial}{\partial x_1} f(x_1, \ldots, x_m) + \ldots + x_m \frac{\partial}{\partial x_m} f(x_1, \ldots, x_m)$$

To illustrate the use of the radial gradient as a tool for automatic differentiation consider

$$P(z_1, z_2) = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} p_{mn} z_1^m z_2^n,$$

and suppose that we want to compute

$$Q(z_1, z_2) = e^{P(z_1,z_2)} = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} q_{mn} z_1^m z_2^n.$$

Then

$$\nabla Q(z_1, z_2)\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \left(z_1 \frac{\partial}{\partial z_1} + z_2 \frac{\partial}{\partial z_2}\right) Q(z_1, z_2) = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}(m+n) q_{mn} z_1^m z_2^n.$$

On the other hand, by applying the chain rule, we have that

$$\begin{aligned}
\nabla Q(z_1, z_2)\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} &= \nabla\left(e^{P(z_1,z_2)}\right)\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \\
&= e^{P(z_1,z_2)} \nabla P(z_1, z_2)\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \\
&= Q(z_1, z_2)\left(z_1 \frac{\partial}{\partial z_1} + z_2 \frac{\partial}{\partial z_2}\right) P(z_1, z_2) \\
&= \left(\sum_{m=0}^{\infty}\sum_{n=0}^{\infty} q_{mn} z_1^m z_2^n\right)\left(\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}(m+n) p_{mn} z_1^m z_2^n\right) \\
&= \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\left(\sum_{j=0}^{m}\sum_{k=0}^{n}(j+k) q_{m-j,n-k} p_{jk}\right) z_1^m z_2^n,
\end{aligned}$$

Matching like powers leads to

$$(m+n) q_{mn} = \sum_{j=0}^{m}\sum_{k=0}^{n}(j+k) q_{m-j,n-k} p_{jk},$$

or

$$q_{mn} = q_{00} p_{mn} + \frac{1}{(m+n)} \sum_{j=0}^{m}\sum_{k=0}^{n} \hat{\delta}_{jk}^{mn} (j+k) q_{m-jn-k} p_{jk},$$

with

$$\hat{\delta}_{jk}^{mn} := \begin{cases} 0 & \text{if } j = 0 \text{ and } k = 0 \\ 0 & \text{if } j = m \text{ and } k = n \\ 1 & \text{otherwise} \end{cases}$$

Given the coefficients of P we compute the coefficients of Q at the cost of a Cauchy product, just as in the one dimensional case.

A similar computation yields the recursion for the Taylor coefficients of the function $P(z_1, z_2)^\alpha$. Indeed, if

$$P(z_1, z_2) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} p_{mn} z_1^m z_2^n,$$

and that we want to compute

$$Q(z_1, z_2) = P(z_1, z_2)^\alpha = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} q_{mn} z_1^m z_2^n.$$

Again, the radial gradient of Q is

$$\nabla Q(z_1, z_2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} (m+n) q_{mn} z_1^m z_2^n.$$

and is also given by

$$\begin{aligned} \nabla Q(z_1, z_2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} &= \nabla P(z_1, z_2)^\alpha \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \\ &= \alpha P(z_1, z_2)^{\alpha - 1} \nabla P(z_1, z_2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}. \end{aligned}$$

Now, multiply both sides by P get

$$P(z_1, z_2) \nabla Q(z_1, z_2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \alpha Q(z_1, z_2) \nabla P(z_1, z_2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

or

$$\left(\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} p_{mn} z_1^m z_2^n \right) \left(\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} (m+n) q_{mn} z_1^m z_2^n \right) = \left(\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \alpha q_{mn} z_1^m z_2^n \right) \left(\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} (m+n) p_{mn} z_1^m z_2^n \right)$$

and taking Cauchy products gives

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \sum_{j=0}^{m} \sum_{k=0}^{n} (j+k) p_{m-j,n-k} q_{jk} z_1^m z_2^n = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \sum_{j=0}^{m} \sum_{k=0}^{n} \alpha (j+k) q_{m-j,n-k} p_{jk} z_1^m z_2^n.$$

Matching like powers we have

$$\sum_{j=0}^{m} \sum_{k=0}^{n} (j+k) p_{m-j,n-k} q_{jk} = \sum_{j=0}^{m} \sum_{k=0}^{n} \alpha (j+k) q_{m-j,n-k} p_{jk},$$

or

$$(m+n)p_{00}q_{mn}+\sum_{j=0}^{m}\sum_{k=0}^{n}\hat{\delta}_{jk}^{mn}(j+k)p_{m-j,n-k}q_{jk} = \alpha(m+n)q_{00}p_{mn}+\sum_{j=0}^{m}\sum_{k=0}^{n}\hat{\delta}_{jk}^{mn}\alpha(j+k)q_{m-j,n-k}p_{jk}.$$

Isolating q_{mn} gives

$$q_{mn} = \alpha p_{00}^{\alpha-1}p_{mn} + \frac{1}{(m+n)p_{00}}\sum_{j=0}^{m}\sum_{k=0}^{n}\hat{\delta}_{jk}^{mn}(j+k)\left(\alpha q_{m-j,n-k}p_{jk} - p_{m-j,n-k}q_{jk}\right).$$

Part V

Appendices

Chapter 19

Appendix

In this chapter we present some definitions, results and notations that are used throughout the book. The results from this chapter can be found in the following references [**?**, **?**, **?**, **?**, **?**, **?**]

19.1 Topological and Metric Spaces

Definition 19.1.1. A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X with the following properties:

(1) $\emptyset$ and X are in $\mathcal{T}$.

(2) The union of the elements of any subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

(3) The intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

A *topological space* is a pair $(X, \mathcal{T})$ where X is a set and $\mathcal{T}$ is a topology on X. We often refer to X as a topological space when there is no need to explicitly indicate the topology $\mathcal{T}$.

Definition 19.1.2. Given a topological space X with topology $\mathcal{T}$ we say that a subset U of X is *open* if U belongs to $\mathcal{T}$. A subset F of X is *closed* if its complement $F^c \stackrel{\text{def}}{=} X \setminus F$ is open. Hence a topological space is a set X together with a collection $\mathcal{T}$ of open subsets of X. Given a point $x \in X$, if $x \in U$ and U is open, we say that U is a *neighborhood* of x.

Proposition 19.1.3. *Let X be a topological space. Then we have the following:*

(1) $\emptyset$ and X are closed.

(2) Arbitrary intersections of closed sets are closed.

(3) Finite unions of closed sets are closed.

Definition 19.1.4. Let A be a subset of a topological space X. We define the *interior* of A, denoted $\operatorname{int}(A)$, as the union of all open sets contained in A, and the *closure* of A, denoted $\overline{A}$, as the intersection of all closed sets containing A. In other words, $\operatorname{int}(A)$ is the largest open set contained in A and $\overline{A}$ is the smallest closed set containing A.

Definition 19.1.5. Let A be a subset of a topological space X. A collection $\mathcal{C}$ of subsets of X is said to *cover* A, or to be a *covering* of A, it the union of the elements of $\mathcal{C}$ contains A. It is called an *open covering* of A if its elements are open subsets of X.

Definition 19.1.6. A subset K of a topological space X is said to be *compact* if every open covering of K contains a finite subcollection that also covers K.

Definition 19.1.7. A topological space X is said to be *locally compact* if for every $x \in X$ there is a compact set K that contains a neighborhood of x.

Definition 19.1.8. A *metric space* is a pair $(X, \mathbf{d})$ where X is a set and $\mathbf{d}$ is a *metric* on X, i.e., a function $\mathbf{d}\colon X \times X \to [0, \infty)$ such that for any $x, y, z \in X$, the following holds

(1) $\mathbf{d}(x, y) = 0$ if and only if $x = y$.

(2) $\mathbf{d}(x, y) = \mathbf{d}(y, x)$.

(3) $\mathbf{d}(x, z) \leq \mathbf{d}(x, y) + \mathbf{d}(y, z)$ (triangle inequality).

The value $\mathbf{d}(x, y)$ is called the *distance* from x to y.

If $(X, \mathbf{d})$ is a metric space, given a point $x_0 \in X$ and an $r > 0$, the *open ball of radius r centered at x_0* is given by

$$B_r(x_0) \stackrel{\text{def}}{=} \{x \in X \mid \mathbf{d}(x_0, x) < r\},$$

and the *closed ball of radius r centered at x_0* is given by

$$\overline{B_r(x_0)} \stackrel{\text{def}}{=} \{x \in X \mid \mathbf{d}(x_0, x) \leq r\}.$$

Definition 19.1.9. Let $(X, \mathbf{d})$ be a metric space and $A \subset X$. We say that A is *open* in X if, for every $x \in A$ there is an $\epsilon > 0$ such that $B_\epsilon(x) \subset A$.

It is easy to see that the collection of all open sets as above define a topology on X. Hence a metric space is a topological space. This topology is called the *metric topology* or the *topology induce by the metric*.

Definition 19.1.10. Let X be a vector space over $\mathbb{R}$ or $\mathbb{C}$. X is a *topological vector space* if with respect to the topology on X vector addition $X \times X \to X$ and scalar multiplication $\mathbb{R} \times X \to X$ ($\mathbb{C} \times X \to X$) are continuous functions.

Definition 19.1.11. Let $(X, \mathbf{d})$ denote a metric space. A function $T\colon X \to X$ is a *contraction* if there is a number $\lambda \in [0, 1)$, called a *contraction constant*, such that

$$\mathbf{d}(T(x), T(y)) \leq \lambda \mathbf{d}(x, y)$$

for all $x, y \in X$.

19.2 Normed Vector Spaces

Throughout this section $\mathbb{F}$ will denote either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$.

Definition 19.2.1. Let X be a vector space over $\mathbb{F}$. A function $\|\cdot\|\colon X \to [0,\infty)$ is a *norm* on X if it satisfies the following conditions:

(1) $\|x\| = 0$ if and only if $x = 0$.

(2) $\|\alpha x\| = |\alpha|\|x\|$ for all vectors $x \in X$ and scalars $\alpha \in \mathbb{F}$.

(3) $\|x+y\| \le \|x\| + \|y\|$ for all vectors $x, y \in X$.

A *normed vector space* is a vector space X equipped with a norm $\|\cdot\|$.

Definition 19.2.2. Let V and W be normed vector spaces with norms denoted by $\|\cdot\|_V$ and $\|\cdot\|_W$, respectively. Let $A\colon V \to W$ be a linear map. The *norm* of A is defined by

$$\|A\| := \sup_{x\neq 0} \frac{\|Ax\|_W}{\|x\|_V} = \sup_{\|x\|_V=1} \|Ax\|_W.$$

Observe that

$$\|Ax\|_W \le \|A\|\|x\|_V.$$

Using different norms on vector spaces results in different norms on operators. It is left to the reader the check that given a matrix $A = \{a_{ij}\}_{i,j}$

$$\begin{array}{cc} \text{Vector Norm} & \text{Matrix Norm} \\ \|x\|_1 := \displaystyle\sum_{k=1}^{n} |x_k| & \|A\|_1 = \displaystyle\max_{1\le j\le n} \sum_{i=1}^{n} |a_{i,j}| \\ \|x\|_2 := \sqrt{\displaystyle\sum_{k=1}^{n} x_k^2} & \|A\|_2 = \sqrt{r_\sigma(A^*A)} \\ \|x\|_\infty := \displaystyle\max_{1\le k\le n} |x_k| & \|A\|_\infty = \displaystyle\max_{1\le i\le n} \sum_{j=1}^{n} |a_{i,j}| \end{array} \tag{19.1}$$

Here $r_\sigma(M)$ denotes the maximum of the magnitudes of the eigenvalues of the matrix M.

Definition 19.2.3. Let X be a linear space. Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ are *equivalent* if there exists constants $K_0, K_1 > 0$ such that

$$K_0\|x\|_a \le \|x\|_b \le K_1\|x\|_a.$$

It is left to the reader to check that this defines an equivalence relation and furthermore that an alternative statement of the definition is that

$$K_0 \leq \|x\|_b \leq K_1.$$

for all $x \in X$ such that $\|x\|_a = 1$.

Proposition 19.2.4. *If X is a finite dimensional linear space, then all norms are equivalent.*

Proof. Assume the dimension of X is N. Let $\{e_n \mid n = 1, \ldots, N\}$ be a basis for X. Then, for every $x \in X$, there are a unique set of coefficients $\{x_n \mid n = 1, \ldots, N\}$ such that

$$x = \sum_{n=1}^{N} x_n e_n.$$

Observe that it is sufficient to prove all norms are equivalent to a single norm. It is left to the reader to check that

$$\|x\|_1 \overset{\text{def}}{=} \sum_{n=1}^{N} |x_n|$$

is a norm. Let $\|\cdot\|$ be an arbitrary norm on X we need to show that there are fixed positive constants K_0 and K_1 such that

$$K_0 \leq \|x\| \leq K_1 \tag{19.2}$$

for all $x \in X$ such that $\|x\|_1 = 1$. Observe that demonstrating (19.2) is equivalent to showing that $\|\cdot\|$ is a bounded function on the unit sphere S^{n-1} (under the $\|\cdot\|_1$ norm).

Since S^{n-1} is compact it is sufficient to show that $\|\cdot\| \colon V \to \mathbb{R}$ is a continuous function under the topology on V induced by $\|\cdot\|_1$. Fix $\epsilon > 0$ and define

$$\delta = \frac{\epsilon}{\max_n \|e_n\|}.$$

Choose $x, y \in X$ such that $\|x - y\|_1 < \delta$. Then

$$|\|x\| - \|y\|| \leq \|x - y\| \leq \sum_{n=1}^{N} |x_n - y_n| \|e_n\| \leq \|x - y\|_1 \max_n \|e_n\| \leq \epsilon.$$

Hence $\|\cdot\| \colon V \to \mathbb{R}$ is a continuous function. □

If X is a normed vector space, it is easy to check that $\mathbf{d}(x, y) = \|x - y\|$ defines a metric on X.

Definition 19.2.5. A *Banach space* is a normed vector space which is complete with respect to the metric defined by the norm.

Definition 19.2.6. Let X be a vector space over $\mathbb{F}$. A map $\langle \cdot, \cdot \rangle : X \times X \to \mathbb{F}$ is an *inner product* (or *scalar product*) on X if for all $x, y, z \in X$ and all $\alpha \in \mathbb{F}$, the following holds:

(1) $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$.

(2) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$.

(3) $\langle x, y \rangle = \overline{\langle y, x \rangle}$.

(4) $\langle x, x \rangle > 0$ if $x \neq 0$.

Here, for $\alpha \in \mathbb{F}$, we use the convention that $\overline{\alpha} = \alpha$ if $\mathbb{F} = \mathbb{R}$ and $\overline{\alpha}$ is the complex conjugate of α if $\mathbb{F} = \mathbb{C}$. Also, if $\alpha \in \mathbb{C}$, the statement $\alpha > 0$ means that $\alpha \in \mathbb{R}$ and α is positive.

A vector space X equipped with an inner product is called and *inner product space*. It is easy to see that an inner product defines a norm on X given by

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

It is left to the reader to check the *Schwarz inequality*

$$| \langle x, y \rangle | \leq \|x\| \|y\|$$

which can be used to prove the triangle inequality for the norm defined above.

Definition 19.2.7. An inner product space H is called a *Hilbert space* if H is a Banach space with respect to the norm defined by its inner product.

Definition 19.2.8. Let $J \subset \mathbb{R}$ be an interval. A *weight* is a non-negative integrable function w defined on J (it may assume the value of infinity) such that if $g \in C^0(J, \mathbb{R})$ and

$$\int_J w(t) g(t) \, dt = 0$$

then $g \equiv 0$.

We are primarily interested in the weights

$$\begin{aligned} w(t) &\equiv 1 & 0 \leq t \leq p \\ w(t) &= \frac{1}{\sqrt{1 - t^2}} & -1 \leq t \leq 1. \end{aligned}$$

Example 19.2.9. *Given $0 < p < \infty$ and $f \colon X \to \mathbb{C}$ a Lebesgue measurable function, define*

$$\|f\|_p \stackrel{\text{def}}{=} \left(\int_X |f|^p \, dx \right)^{1/p}$$

and let $L^p(X)$ be defined to be the set of all measurable functions $f \colon X \to \mathbb{C}$ for which $\|f\|_p < \infty$. If we identify functions which differ on a set of measure 0 then $L^p(X)$ is a Banach space.

As the following example indicates there are different inner products that can be imposed on $L^2([a,b],\mathbb{R})$.

Example 19.2.10. The Banach space $L^2([a,b],\mathbb{R})$ with inner product

$$(f,g) \stackrel{\text{def}}{=} \int_a^b \frac{f(t)g(t)}{\sqrt{1-t^2}}\,dt$$

is a Hilbert space

19.3 Calculus

Second order Taylor expansion.

Theorem 19.3.1. *Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be differentiable at $(\lambda_0, x_0) \in \mathbb{R}$. Then there exists a function $R_{(\lambda_0,x_0)}\colon \mathbb{R}^2 \to \mathbb{R}$ so the*

$$f(\lambda,x) = f(\lambda_0,x_0) + \nabla f(\lambda_0,x_0)\begin{bmatrix} \lambda-\lambda_0 \\ x-x_0 \end{bmatrix} + R_{(\lambda_0,x_0)}(\lambda-\lambda_0, x-x_0),$$

and

$$\lim_{(\lambda,x)\to(\lambda_0,x_0)} R_{(\lambda_0,x_0)}(\lambda-\lambda_0, x-x_0).$$

Moreover, if f is twice continuously differentiable on $D := [\lambda_0-r_*, \lambda_0+r_*]\times[x_0-r_*, x_0+r_*]$ then

$$\sup_{(\lambda,x)\in D} \left| R_{(\lambda_0,x_0)}(\lambda-\lambda_0, x-x_0)\right| \leq 2Mr_*^2,$$

where $M > 0$ is a bound of the form

$$\max\left(\sup_{(\lambda,x)\in D}\left|\frac{\partial^2}{\partial\lambda^2} f(\lambda,x)\right|, \sup_{(\lambda,x)\in D}\left|\frac{\partial^2}{\partial\lambda\partial x} f(\lambda,x)\right|, \sup_{(\lambda,x)\in D}\left|\frac{\partial^2}{\partial x^2} f(\lambda,x)\right|\right) \leq M.$$

The Inverse Function theorem.

Theorem 19.3.2. *Let $U \subset \mathbb{R}^n$ be an open set. Let $f\colon U \to \mathbb{R}^n$ be a C^k function with $k \geq 1$. If $Df(\hat{x})$ is invertible at a point $\hat{x} \in U$, then there exits an open neighborhood $V \subset U$ of $\hat{x}$ such that $f\colon V \to f(V)$ is invertible with C^k inverse.*

Theorem 19.3.3. *Let $U \subset \mathbb{R}^n$ be any open set. There exists a function $f \in C^\infty(\mathbb{R}^n)$ such that $f(x) > 0$ for all $x \in U$, while $f(x) = 0$ for all $x \in \mathbb{R}^n \setminus U$.*

Proof. First consider the special case that $U = B_r(x_0) = \{x \in \mathbb{R} \mid \|x - x_0\|_2 < r\}$. Then we take

$$f_{x_0,r}(x) \overset{\text{def}}{=} \begin{cases} \exp\left(\frac{1}{\|x-x_0\|_2^2 - r^2}\right) & \text{for} \|x - x_0\|_2 < r \\ 0 & \text{for} \|x - x_0\|_2 \geq r. \end{cases}$$

For the general case we write U as a countable union of open balls:

$$U = \bigcup_{n \in \mathbb{N}} B_{r_n}(x_n).$$

We then choose

$$f(x) \overset{\text{def}}{=} \sum_{n=0}^{\infty} a_n \, f_{x_n,r_n}(x),$$

with $a_n > 0$ decreasing sufficiently fast. To be specific, let $f_n := f_{x_n,r_n}$, then we introduce the following bounds on their derivatives:

$$M_n \overset{\text{def}}{=} \max\left\{1, \max_{|\alpha| \leq n} \max_{x \in \mathbb{R}^n} |\partial^\alpha f_n(x)|\right\}.$$

We now choose

$$a_n \overset{\text{def}}{=} \frac{1}{M_n 2^n}.$$

For any $\alpha \in \mathbb{N}^d$ we can show that $\sum_{n=0}^{\infty} a_n \partial^\alpha f_n$ converges uniformly on $\mathbb{R}^n$. Namely, for all $n \geq |\alpha|$ we can estimate

$$\max_{x \in \mathbb{R}^n} |a_n \partial^\alpha f_n(x)| \leq |a_n| M_n = 2^{-n},$$

and the result follows from the dominated convergence theorem. We conclude that $f \in C^\infty(\mathbb{R}^n)$. Since $a_n > 0$ we have $f(x) > 0$ for all $x \in U$, whereas f vanishes outside U. $\square$

We note that it is not difficult to see that in case U is unbounded (and trivially for bounded U) the function $f(x)$ constructed in the proof tends to zero as $x \to \infty$, as do all of its derivatives.

19.4 Analytic Functions

Here we will define analytic functions and introduce the multi-index notation. (Still working on this)

The proof of this theorem is analogous to that of the one-dimensional ODE, however the higher dimensionality leads to the need for more sophisticated notation. An n-tuple of nonnegative integers $\alpha = (\alpha_1, \ldots, \alpha_n)$ is called a *multi-index*. We define

$$|\alpha| \overset{\text{def}}{=} \sum_{k=1}^{n} \alpha_k \quad \text{and} \quad \alpha! \overset{\text{def}}{=} \alpha_1! \alpha_2! \cdots \alpha_n!.$$

We extend this notation to vector valued functions. Given $x \in \mathbb{R}^n$

$$x^\alpha \stackrel{\text{def}}{=} x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

As an exercise in use of multi-index notation we leave it to the reader to check that

$$(x_1 + \ldots + x_n)^k = \sum_{|\alpha|=k} \frac{|\alpha|!}{\alpha!} x^\alpha. \tag{19.3}$$

We use multi-indices to express higher order derivatives, in particular

$$\partial^\alpha \stackrel{\text{def}}{=} \left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \left(\frac{\partial}{\partial x_2}\right)^{\alpha_2} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}.$$

19.5 Taylor expansion of an analytic function

Let $z_0 \in \mathbb{C}$ and $r > 0$. Then we can write $z = x + iy$ for some $x, y \in \mathbb{R}$. Define the complex absolute value $|z| = \sqrt{x^2 + y^2}$. Define the open disk of radius r about z_0 by

$$D_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\}.$$

Let $\Omega \subset \mathbb{C}$ be an open set and $f\colon \Omega \to \mathbb{C}$ be a complex valued function. The function f is called *holomorphic* if for all $z_0 \in \Omega$ the complex derivative

$$f'(z_0) = \frac{d}{dz} f(z_0) := \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0},$$

exists and is finite.

As one learns in a standard course in complex analysis, a function is holomorphic if and only if it can be represented by convergent power series. The power series representation of an analytic function plays a key role in the next two chapters, and we will make the preceding statement precise. First we must recall a few basic facts about power series.

Theorem 19.5.1. *Let $\{a_n\}_{n=0}^\infty \subset \mathbb{C}$ be a sequence of complex numbers. Then there exists a $0 \le R \le \infty$ so that the power series $\sum_{n=0}^\infty a_n z^n$ converges absolutely if $|z| < R$ and diverges if $|z| > R$. With the convention that $1/0 = \infty$ and $1/\infty = 0$, R is given by Hadamard's formula*

$$\frac{1}{R} = \limsup |a_n|^{1/n}.$$

The number R is called the *radius of convergence* of the power series. A proof is found in (CITE RUDIN). Note that the theorem says nothing about the behavior of the series for $|z| = R$, which can be quite complicated.

We now have the first of two important facts: that power series define holomorphic functions.

Theorem 19.5.2. *Let $\{a_n\}_{n=0}^{\infty} \subset \mathbb{C}$ be a sequence of complex numbers and suppose that $R = 1/\limsup |a_n|^{1/n} > 0$. Then $f\colon D_R(0) \to \mathbb{C}$ given by*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n,$$

defines a holomorphic function on $D_R(0)$. Moreover the derivative of f is holomorphic on $D_R(0)$ and has the power series

$$f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1},$$

with the same radius of convergence R.

See (Rudin? Stein? Alhfors?) for the proof. As a corollary we have that the function defined by a power series is infinitely complex differentiable, that the power series of each of the derivatives is given by "term-by-term" differentiation, and that each derivative has radius of convergence $R > 0$. All of the above comments apply to power series centered at points other than zero, i.e. series of the form $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges, along with all its derivatives, on a disk of the form $D_R(z_0)$ with $0 \le R \le \infty$.

Let $\Omega \subset \mathbb{C}$ be an open set. A function $f\colon \Omega \to \mathbb{C}$ is said to be *analytic at* $z_0 \in \Omega$ if there is a sequence of complex numbers $\{a_n\}_{n=0}^{\infty}$ with $R = 1/\limsup |a_n|^{1/n} > 0$ so that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

i.e. f is analytic at $z_0 \in \Omega$ if f has a convergent power series representation in some neighborhood of z_0. A function f is said to be *analytic on* Ω if it is analytic at each of its points. Theorem 19.5.2 says that every analytic function is holomorphic. It is one of the deep facts of complex analysis that the converse is true.

Theorem 19.5.3. *Suppose that f is holomorphic in an open set $\Omega \subset \mathbb{C}$, and that $z_0 \in \Omega$ and $R > 0$ are such that*

$$\overline{D_R(z_0)} \subset \Omega.$$

Define the sequence of complex numbers $\{a_n\}_{n=0}^{\infty} \subset \mathbb{C}$ by the integrals

$$a_n = \frac{1}{2\pi i} \int_{|z-z_0|=R} \frac{f(z)}{(z - z_0)^{n+1}}\, dz, \tag{19.4}$$

where the curve $|z - z_0| = R$ is parameterized so as to have winding number $+1$ aboud z_0. Then f has power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

about z_0, with radius of convergence at least R. In other words, f is analytic at z_0.

For the proof see (ADD CITATIONS). Taken together Theorems 19.5.2 and 19.5.3 show that holomorphic and analytic are equivalent notions, hence the words are sometimes used interchangeably. In the remainder of the book we favor the term analytic. Recall that differentiable implies continuous, so that holomorphic functions are continuous. Since a function continuous on the compact set $\overline{D_R(z_0)}$ attains its maximum and minimum, the integrals defined in Equation 19.4 exist and the numbers a_n are well defined. It is of great importance that the Theorem tells us how to find the power series expansion of an analytic function. Recall also that the Taylor coefficients can be expressed equivalently as

$$a_n = \frac{1}{n!}\frac{d^n f}{dz^n}(z_0) = \frac{f^{(n)}(z_0)}{n!},$$

and that the Taylor coefficients uniquely determine an analytic function. In particular, $f(z) = 0$ for all $z \in \Omega$ if and only if $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ with $a_n = 0$ for all $n \geq 0$.

We also exploit the fact that the Taylor coefficients of a bounded analytic function satisfy certain exponential growth rates. This claim is made precise in the following Theorem.

Theorem 19.5.4 (Cauchy Estimates). *Suppose that f is analytic on an open set $\Omega \subset \mathbb{C}$. Suppose that $z_0 \in \Omega$ and $R > 0$ are such that*

$$\overline{D_R(z_0)} \subset \Omega.$$

Then

$$|a_n| \leq \frac{1}{R^n} \sup_{z \in D_R(z_0)} |f(z)|.$$

Remark 19.5.5. It follows that if f is bounded on $D_R(z_0)$ and $0 < \nu < R$. Let $0 < M < \infty$ have that $|f(z)| \leq M$ for all $z \in D_R(z_0)$. Then

$$\begin{aligned} \sum_{n=0}^{\infty} |a_n|\nu^n &= \sum_{n=0}^{\infty} |a_n| R^n \left(\frac{\nu}{R}\right)^n \\ &\leq \sum_{n=0}^{\infty} M \left(\frac{\nu}{R}\right)^n \\ &= \frac{M}{1 - \frac{\nu}{R}} < \infty. \end{aligned}$$

So certain weighted sums of the Taylor coefficients of an analytic function converge. This fact plays a critical role in the next two chapters.

Remark 19.5.6 (The Cauchy product formula). Suppose that $z_0 \in \mathbb{C}$ and that $f, g\colon D_r(z_0) \to \mathbb{C}$ are analytic for some $r > 0$. Let

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n, \qquad \text{and} \qquad g(z) = \sum_{n=0}^{\infty} b_n(z - z_0)^n,$$

denote the power series expansions at z_0. It follows from the product rule for differentiation that the product $f \cdot g$ is analytic on $D_r(z_0)$. By Merten's theorem and the fact that the power series converge absolutely for any $z \in D_r(z_0)$, we have that the power series for the product is given by

$$(f \cdot g)(z) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_{n-k} b_k \right) (z - z_0)^n, \tag{19.5}$$

and that this series has radius of convergence at least $r > 0$.

It is natural to define certain Banach spaces of analytic functions. Let $\Omega \subset \mathbb{C}$ be an open set. Given $f : \Omega \to \mathbb{C}$, write

$$\|f\|_{C^0(\Omega)} := \sup_{z \in \Omega} |f(z)|,$$

to denote the "C^0 norm" or "norm of uniform convergence" on Ω. The key fact is that if $f, f_n \colon \Omega \to \mathbb{C}$ are functions with the $\{f_n\}_{n=0}^{\infty}$ analytic, and if

$$\lim_{n \to \infty} \|f_n - f\|_{C^0(\Omega)} = 0,$$

then f is analytic on Ω. From this we have that the vector space

$$C^{\omega}(\Omega, \mathbb{C}) := \left\{ f \colon \Omega \to \mathbb{C} : f \text{ is analytic and } \|f\|_{C^0(\Omega)} < \infty \right\}.$$

is a Banach space. It is, furthermore a Banach algebra with product given by pointwise multiplication of functions. The space $C^{\omega}(D_r(z_0), \mathbb{C})$ is referred to as *the disk algebra*.

19.6 Regularity of Solutions

In order to focus on a particular framework for constructive proofs for ODEs, we restrict our attention to analytic vector fields. As we show below, solutions of initial value problems for analytic vector fields are themselves analytic. This fact allows us to restrict our attention to orbit segments with convergent power series expansions, and this restriction colors the analysis in the remainder of the chapter.

Definition 19.6.1. Let $U \subset \mathbb{C}$ be an open set and suppose that $f \colon U \to \mathbb{C}$ is analytic. Suppose that $\gamma \colon [a, b] \to \mathbb{C}$ is a continuous and piecewise differentiable curve. More precisely suppose that

$$[a, b] = \cup_{j=1}^{N} [t_{j-1}, t_j],$$

with $a = t_0 < t_1 < \ldots < t_{N-1} < t_N = b$, and that each $\gamma|_{[t_{j-1}, t_j]}$ is differentiable on (t_{j-1}, t_j). The *line integral of f over γ* is defined to be

$$\int_{\gamma} f(z)\, dz = \sum_{j=1}^{N} \int_{t_{j-1}}^{t_j} f(\gamma(t)) \gamma'(t)\, dt.$$

Suppose that $\|f(z)\| \leq M$ for all $z \in U$ and that γ is a curve of length L. A key estimate is that

$$\left| \int_\gamma f(z)\, dz \right| \leq ML. \tag{19.6}$$

A deep theorem from complex analysis is the Cauchy integral theorem, which we state now for completeness.

Theorem 19.6.2. *Suppose that $U \subset \mathbb{C}$ is a simply connected open set, and that $\gamma\colon [a,b] \to \mathbb{C}$ is a piecewise smooth, closed curve in U. Then*

$$\int_\gamma f(z)\, dz = 0.$$

It follows that if U is open and simply connected, $z_0, z \in U$, and $\gamma\colon [a,b] \to \mathbb{C}$ is a piecewise smooth curve in U connecting z_0 to z (i.e. a curve with $\gamma(a) = z_0$ and $\gamma(b) = z$) then the path integral $\int_\gamma f(w)\, dw$ depends only on the end points z_0 and z. This fact follows from Theorem 19.6.2 as soon as we observe that if γ_1 and γ_2 are any two curves connecting z_0 to z in U, then $\gamma_1 - \gamma_2$ is a simple closed curve in U. Since the integral depends only on the end points we will write

$$\int_\gamma f(w)\, dw = \int_{z_0}^{z} f(w)\, dw.$$

Definition 19.6.3. Suppose that $U \subset \mathbb{C}$ is an open set and that $f\colon U \to \mathbb{C}$ is an analytic function. A function $F\colon U \to \mathbb{C}$ is a *holomorphic antiderivative* of f on U if

$$\frac{d}{dz} F(z) = f(z),$$

for all $z \in U$.

We can now state the fundamental theorem of calculus for analytic functions.

Theorem 19.6.4 (Complex analytic fundamental theorem of calculus: part I). *Suppose that $U \subset \mathbb{C}$ is a simply connected open set and that $f\colon U \to \mathbb{C}$ is analytic. Choose any $z_0 \in U$, and let γ be a curve connecting z_0 to z in U. Then, the function $F\colon U \to \mathbb{C}$ defined by*

$$F(z) = \int_{z_0}^{z} f(w)\, dw,$$

is the unique antiderivative of f with $F(z_0) = 0$.

Conversely one has:

Theorem 19.6.5 (Complex analytic fundamental theorem of calculus: part II). *Suppose that $U \subset \mathbb{C}$ is an open set and that $f\colon U \to \mathbb{C}$ is an analytic function. Let $F\colon U \to \mathbb{C}$ be a holomorphic antiderivative of f on U. Then for any piecewise smooth curve $\gamma\colon [a,b] \to \mathbb{C}$ with $\gamma([a,b]) \subset U$ on has that*

$$\int_\gamma f(z)dz = F(\gamma(b)) - F(\gamma(a)).$$

Of course these comments apply component wise to functions $f\colon U \to \mathbb{C}^n$ where U is a simply connected open set in $\mathbb{C}$.

Now we are interested in the complex initial value problem

$$\frac{d}{dz}x(z) = f(x(z)), \qquad\qquad x(0) = x_0,$$

with $f\colon U \to \mathbb{C}^n$ an analytic vector field defined on the open set $U \subset \mathbb{C}^n$, and $x_0 \in U$. Let $r > 0$ and define

$$B_r(x_0) = \{x \in \mathbb{C}^n \colon \|x - x_0\| < r\},$$

and

$$D_r = \{z \in \mathbb{C}\colon \|z\| < r\}.$$

Combining the above observations, and noting that $B_r(x_0)$ is open and simply connected, we have the following generalization of Lemma 6.1.1 from Chapter 6.

Lemma 19.6.6. *Assume that $f\colon U \to \mathbb{C}^n$ is an analytic function defined on an open set $U \subset \mathbb{C}^n$, and let $x_0 \in U$. A continuous function $\varphi\colon D_r \to \mathbb{C}^n$ solves the initial value problem $\dot{x} = f(x)$, $x(0) = x_0$ on D_r if and only if*

$$\varphi(z) = x_0 + \int_0^z f(\varphi(w))\, dw.$$

The following generalizes the existence and uniqueness result of Theorem 6.1.5 to the analytic setting.

Proposition 19.6.7. *If $f\colon U \to \mathbb{C}^n$ is an analytic function defined on an open set $U \subset \mathbb{C}^n$ and $x_0 \in U$, then there exists an $r > 0$ and a solution $\varphi\colon D_r \to U$ to the initial value problem*

$$\frac{d}{dz}x(z) = f(x(z)), \quad x(0) = x_0. \tag{19.7}$$

Moreover the solution $\varphi(z)$ is analytic on D_r.

Proof. The vector field f locally Lipschitz on U as analytic implies continuously differentiable. Then the first part of the proof proceeds almost exactly as in the proof of

Theorem 6.1.5, i.e. we choose $\epsilon > 0$ so that $\overline{B_\epsilon(x_0)} \subset U$ and have that there are constants $K, M > 0$ so that

$$\|f(x) - f(y)\| \le K\|x - y\| \quad \text{and} \quad \|f(x)\| \le M,$$

for all $x, y \in \overline{B_\epsilon(x_0)}$. Choose $r > 0$ with

$$r < \min\left\{\frac{\epsilon}{M}, \frac{1}{K}\right\},$$

and define the complete metric space $X \subset C^\omega(D_r, \mathbb{C}^n)$ by

$$X := \overline{B_{Mr}(x_0)} = \left\{\alpha\colon D_r \to \mathbb{C}^n : \alpha \text{ is analytic and } \|\alpha(z) - x_0\|_{C^0(D_r)} \le Mr\right\},$$

where $\|\alpha(z) - x_0\|_{C^0(D_r)} = \sup_{z \in D_r} \|\alpha(z) - x_0\|$. We now look for a fixed point of the integral operator

$$T(\varphi(z)) = x_0 + \int_0^z f(\varphi(w))\, dw,$$

on X, where we take the line integral over the straight line path from 0 to z in D_r. Now estimating as in the proof of Theorem 6.1.5, and employing the bound of Equation (19.6), leads to

$$\|T(\alpha)(z) - x_0\|_{C^0(D_r)} \le Mr,$$

and

$$\|T(\alpha) - T(\beta)\|_{C^0(D_r)} \le Kr\|\alpha - \beta\|_{C^0(D_r)}.$$

Then by the contraction mapping theorem T has a unique fixed point $\varphi \in X$.

To see that $\varphi\colon D_r \to \mathbb{C}^n$ is analytic we proceed as follows. Note that the constant function $x_0 \in X$ is an analytic function. Consider the Picard sequence starting with the function x_0. The functions

$$x_{n+1}(z) = x_0 + \int_0^z f(x_n(w))\, dw,$$

are each analytic, as the composition of analytic functions is analytic and the line integral defines an analytic function by Theorem 19.6.4. Now since $x_{n+1}(z)$ converge uniformly (i.e. in the C^0 norm) to $\varphi(z)$, it follows that φ is analytic. □

Example 19.6.8. Direct substitution shows that the unique solution to the IVP

$$\dot{y}_k = g_k(y) = \frac{Cr}{r - \sum_{j=1}^n y_j}, \quad y(0) = 0, \quad k = 1, \ldots, n,$$

where C, $r > 0$, is given by

$$y_k(t) = \frac{r}{n} - \sqrt{\left(\frac{r}{n}\right)^2 - \frac{2Cr}{n} t},$$

and this is an analytic function for $|t| < \frac{r}{2Cn}$.

Alternative proof of Proposition 19.6.7. Since f is analytic, we know that the solution x to the IVP is C^∞. Repeated differentiation produces

$$\begin{aligned}
\dot{x}_j(t) &= f_j(x(t)) \\
\ddot{x}_j(t) &= \sum_{m=1}^{n} \frac{\partial f_j}{\partial x_m}(x(t))\dot{x}_m(t) \\
&\vdots \\
x_j^{(k)}(t) &= q_k\left(\{\partial^\alpha f_j(x(t))\}_{|\alpha|<k}, \left\{x_m^{(\ell)}(t)\right\}_{\ell<k, 1\le m\le n}\right)
\end{aligned}$$

where q_k is a polynomial with positive integer coefficients. By assumption $x(0) = 0$, thus

$$x_j^{(k)}(0) = q_k\left(\{\partial^\alpha f_j(0)\}_{|\alpha|<k}, \left\{x_m^{(\ell)}(0)\right\}_{\ell<k, 1\le m\le n}\right).$$

Applying this formula by induction on k, we can eliminate all the lower derivatives $x_m^{(\ell)}(0)$ from the right hand side to get

$$x_j^{(k)}(0) = p_k\left(\{\partial^\alpha f_j(0)\}_{|\alpha|<k}\right),$$

where p_k is again a polynomial with positive integer coefficients.

Using the fact that p_k has positive coefficients, and then Lemma **??**, we obtain

$$\begin{aligned}
x_j^{(k)}(0) &= p_k\left(\{\partial^\alpha f_j(0)\}_{|\alpha|<k}\right) \\
&\le p_k\left(\{|\partial^\alpha f_j(0)|\}_{|\alpha|<k}\right) \\
&\le p_k\left(\{\partial^\alpha g_j(0)\}_{|\alpha|<k}\right) \\
&= y_j^{(k)}(0),
\end{aligned}$$

if g and y are chosen as in Example 19.6.8. Since y is analytic it follows, as in the proof of Theorem **??**, that x is analytic. □

19.7 Eigenvalues of Matrices

Theorem 19.7.1. *Let $A \in M_n(\mathbb{R})$. The eigenvalues of A are the roots of the characteristic polynomial*

$$p(x) = \det(xI - A).$$

Theorem 19.7.2. *Consider the n-th degree polynomials*

$$p(z) = z^n + \sum_{k=0}^{n-1} a_k z^k \quad \textit{and} \quad q(z) = z^n + \sum_{k=0}^{n-1} q_k z^k.$$

Assume that p has m distinct roots $\{\lambda_i \mid i = 1, \ldots, m\}$ with multiplicities α_i, respectively. Choose $\epsilon > 0$ such that $\overline{B_\epsilon(\lambda_i)}$, $i = 1, \ldots, m$ are disjoint. Then, there exists $\delta > 0$ such that if $|a_k - b_k| < \delta$, $k = 0, \ldots, n-1$, then q has exactly α_i roots (counting mulitplicities) in $\overline{B_\epsilon(\lambda_i)}$.

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